

Pharmacology: Antineoplastic Agents Part 1

Nonselective (conventional)

Cytotoxic Chemotherapy

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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How will knowledge of cancer drugs be relevant for my future practice as a generalist or a specialist?

Primary care physicians – indeed, all health care professionals – are members of the cancer survivors health care team.

Following this session, students should be able to explain the mechanisms of action, resistance, general adverse effects, dose-limiting toxicities, cancer uses, and drug interactions:

1. Explain the fundamental principles of nonselective cytotoxic chemotherapy on inducing DNA damage in relation to cell cycle phases and kinetics.
2. Relate the anticancer mechanisms actions of each category of antineoplastic drugs to their effects.
3. Correlate the mechanisms of resistance with the mechanisms of drug action and cellular mechanisms of detoxification and reduced drug accumulation.
4. Identify the dose-limiting toxicities of the cytotoxic chemotherapies.
5. Apply the mechanistic and pharmacokinetic properties of the anticancer drugs to their potential adverse effects and drug interactions when presented with a case vignette.
6. Create a concept map or other technique integrating the mechanisms of action and resistance, potential toxicities, drug-disease and drug-drug interactions to the selection of drugs for combination antineoplastic therapeutic regimens.

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Optional materials:

- Video Lecture | Dr. Goldstein's Notes handout | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Scholar Rx Bricks:

Name of Discipline/ Organ System: General Pharmacology > Chemotherapy > Chemotherapy Agents

Title of Brick: Alkylating and Platinum Agents

Link: https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0016

Title of Brick: Antimetabolite Agents

Link: https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0017

Title of Brick: Topoisomerase Inhibitors, Microtubule Inhibitors, and Antitumor Antibiotics

Link: https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0019

Access Medicine Katzung Basic & Clinical Pharmacology, 16e, 2024; Chapter 54: Cancer Chemotherapy

<https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/book.aspx?bookid=3382#281742356>

THE REVIEW BOOKS BELOW HAVE PRACTICE QUESTIONS

LWW Health Library, Medical Education; Lippincott's Illustrated Reviews, 8e, 2023; Chapter 37: Anticancer Drugs

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329619&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 54: Cancer Chemotherapy

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285600220>

Chemotherapy:

Cytotoxic chemicals that kill cells or stop cell division leading to apoptosis

when proteins that initiate apoptosis, such as p53, are functional.

Alkylating agents All phases of cell cycle	Nucleophilic attack on DNA → DNA cross-linking → DNA damage → activate DNA mismatch repair (MMR) pathway → DNA strand breaks by repair enzymes → apoptosis.
Platinum analogs All phases of cell cycle	Form covalent metal adducts with DNA → DNA cross-linking → DNA damage → activates MMR ultimately leading to apoptosis
Antimetabolites S phase phase	Folate, purine, or pyrimidine analogs that inhibit DNA synthesis and repair in proliferating cells → p53 activation → apoptosis.
Antitumor antibiotics G2/M phase primarily	Intercalate in DNA, inhibit topoisomerase II, and generates free radicals → irreparable strand breaks, oxidative damage to DNA, proteins, and cell membranes.
Antimitotics M phase	Inhibit polymerization in microtubules (vinca alkaloids) or stabilize microtubules and prevent depolymerization (taxanes) → suppress spindle microtubule function → cell cycle arrest → apoptosis
Camptothecin analogs S phase	Form drug-DNA-topoisomerase I complex → prevents resealing step → DNA damage accumulates → apoptosis
Epipodophyllotoxins late S and G2 phase	Form drug-DNA-topoisomerase II complex → irreparable strand breaks → DNA damage accumulates → apoptosis

Biologic and small molecule targeted agents:

These drugs block specific molecules in oncogenic pathways that lead to dysregulated cancer cell growth and survival interferes with growth, progression, and spread of cancer.

- Signal transduction inhibitors
 - Inhibitors of growth factors and receptors
 - Inhibitors of intracellular pathways
- Angiogenesis inhibitors
- Antibody-drug conjugates
- Immune checkpoint inhibitors
- CAR-T cells

Hormonal anticancer therapeutics:

The growth of some cancers is regulated by or dependent on hormones.

- Glucocorticoids (antiproliferative and lympholytic properties)
- Estrogen receptor antagonists
- Progesterone receptor agonists (antiproliferative effects)
- Androgen receptor antagonists
- Steroidogenesis inhibitors
- Gonadotropin releasing hormone (GnRH) agonists and antagonists

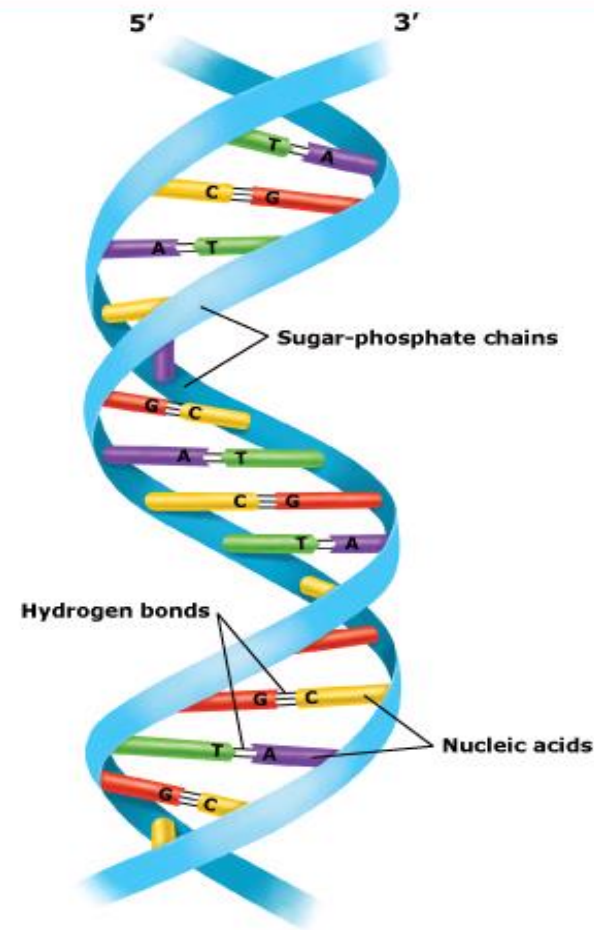
The hormonal drugs are not included in this lecture.

General Overview of Concepts

- **What are the goals of cancer therapy?**
 - To rid the body of malignant cells
 - If cure is not possible, then the goal is to shrink the tumors and relieve symptoms.
- **Antineoplastic therapy entails surgery, radiation, anticancer drugs**

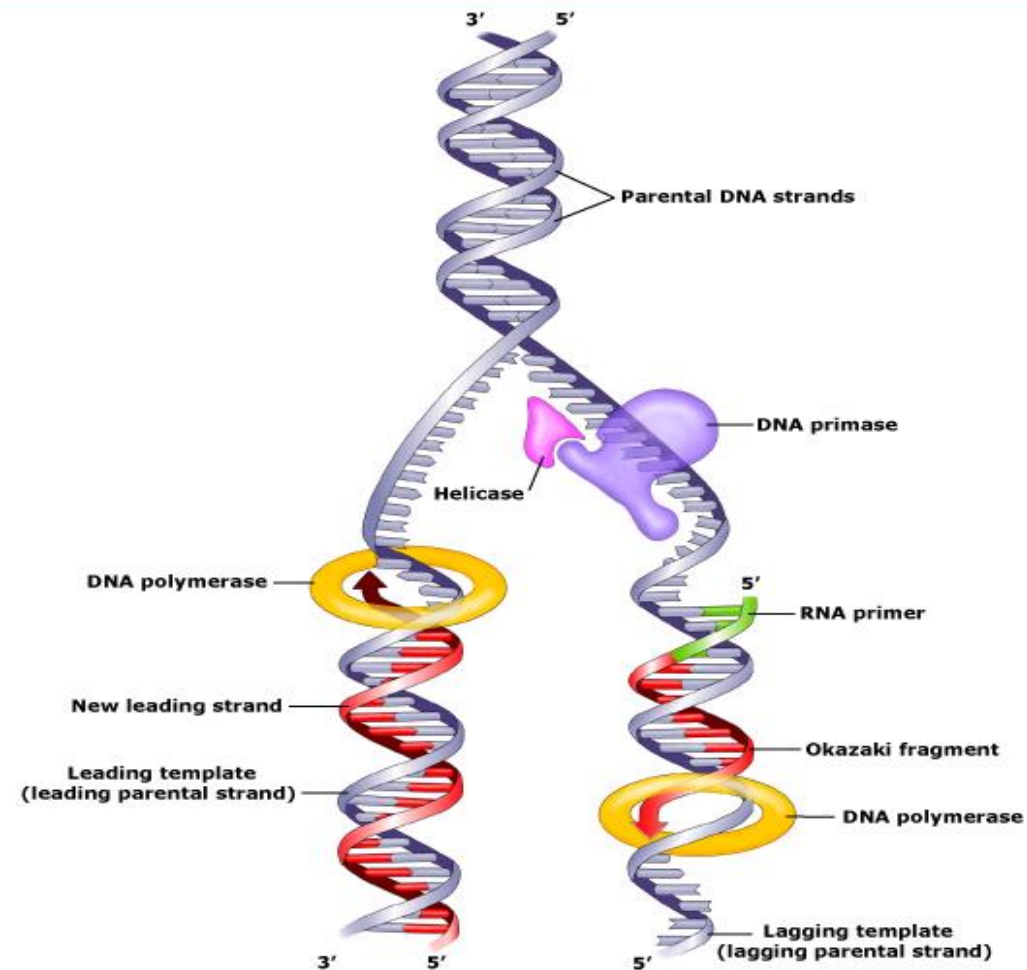
What are the patient's goals?

Structure of DNA

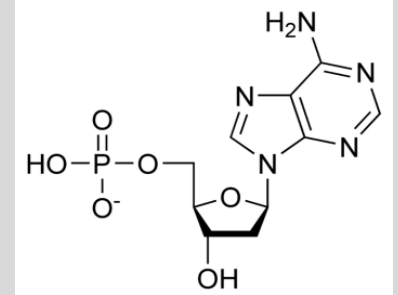


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DNA replication



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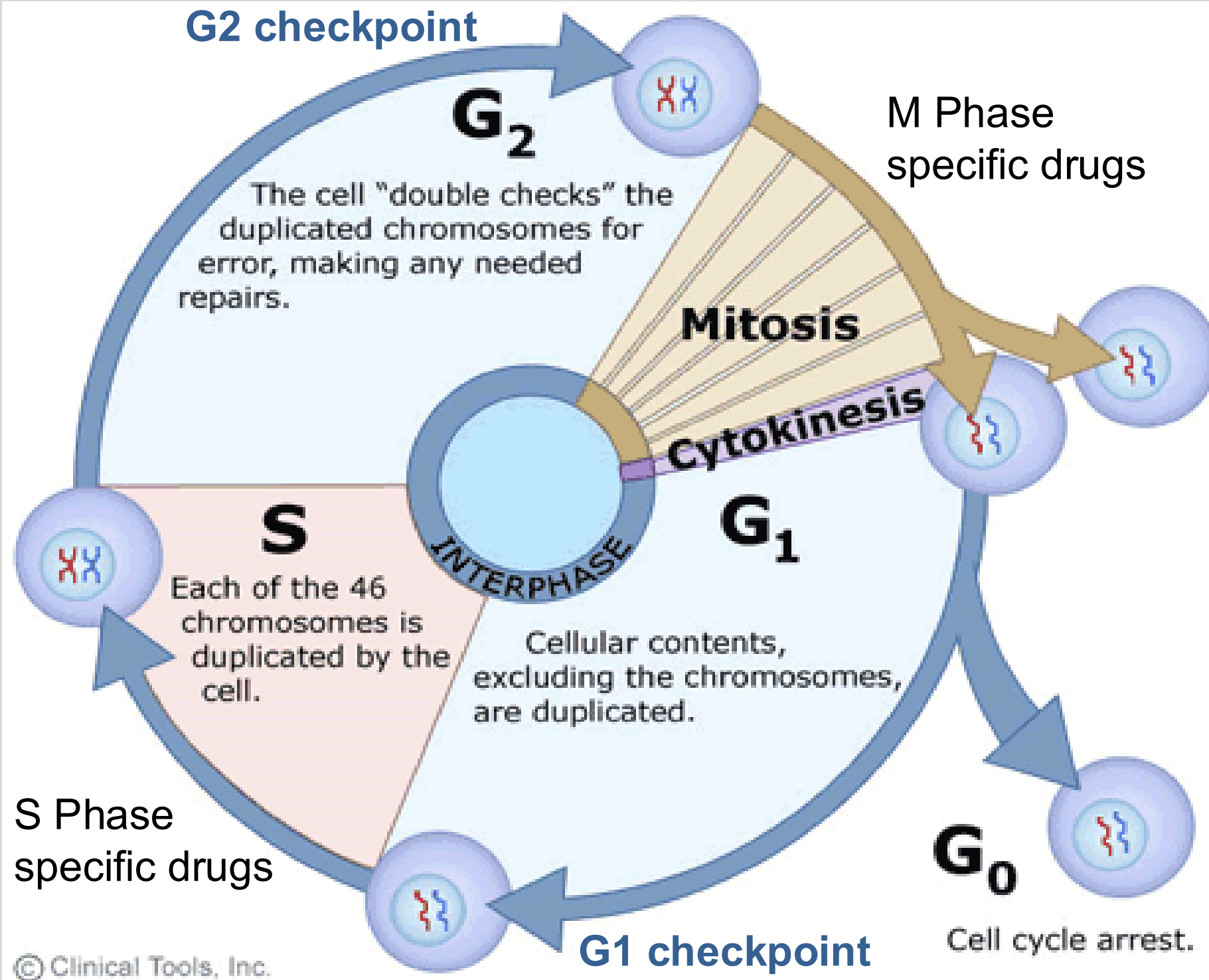
Deoxynucleotide
(deoxyadenosine
monophosphate)

Helicase: unwinds DNA

DNA primase: creates primer (which is RNA that is later removed by exonuclease)

DNA polymerase: assembles the nucleotides, adding them to the previous 3'-OH group

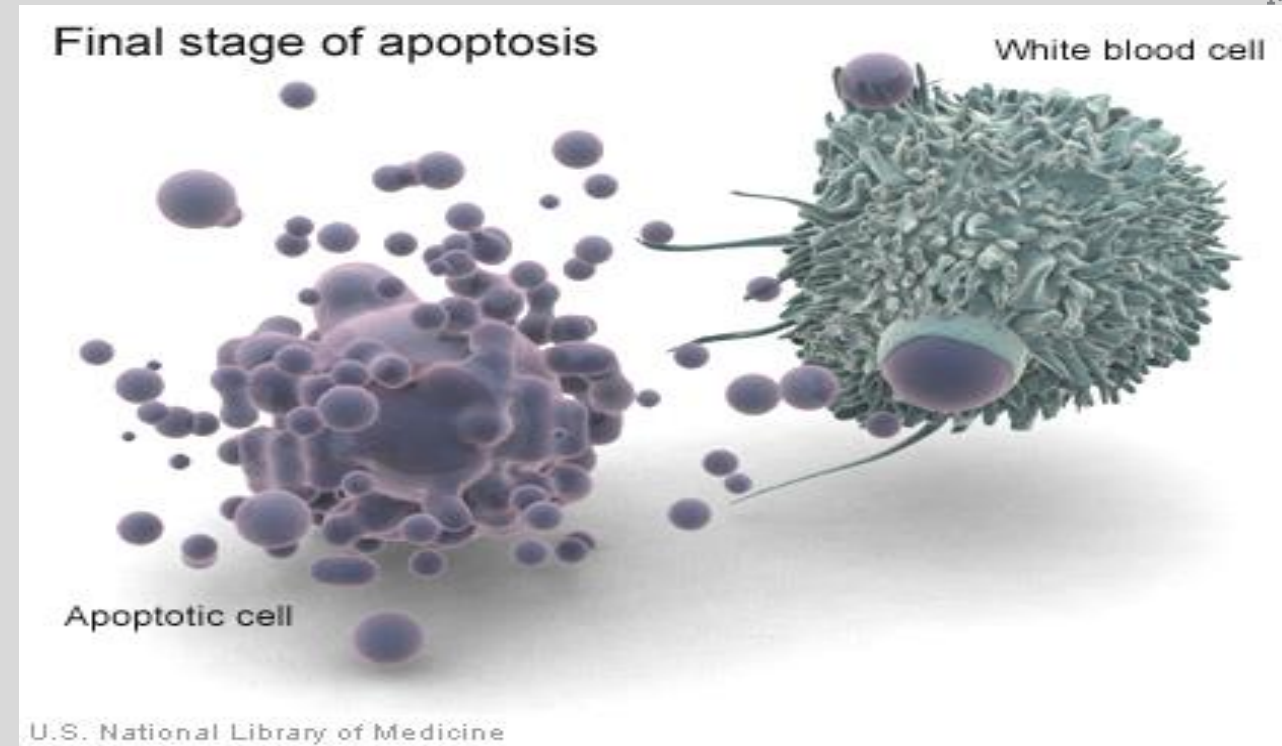
Okazaki fragment: newly synthesized DNA formed on the lagging template strand



- Checkpoints detect DNA damage leading to repair or, if not reparable, to apoptosis.
- Tumors with a high percentage of proliferating cells are most responsive to chemotherapy.
- Slow-growing tumors often respond to chemotherapy that inflicts high-level DNA damage that overwhelms the tumor's repair mechanisms.

Anticancer agents are intended to trigger apoptosis

- DNA alkylation or adduct (phase nonspecific)
- DNA synthesis inhibitors (S phase actions)
- Cell division inhibition (M phase actions)
- Targeted therapy
 - Block cancer growth pathways
 - Enhance T cell function

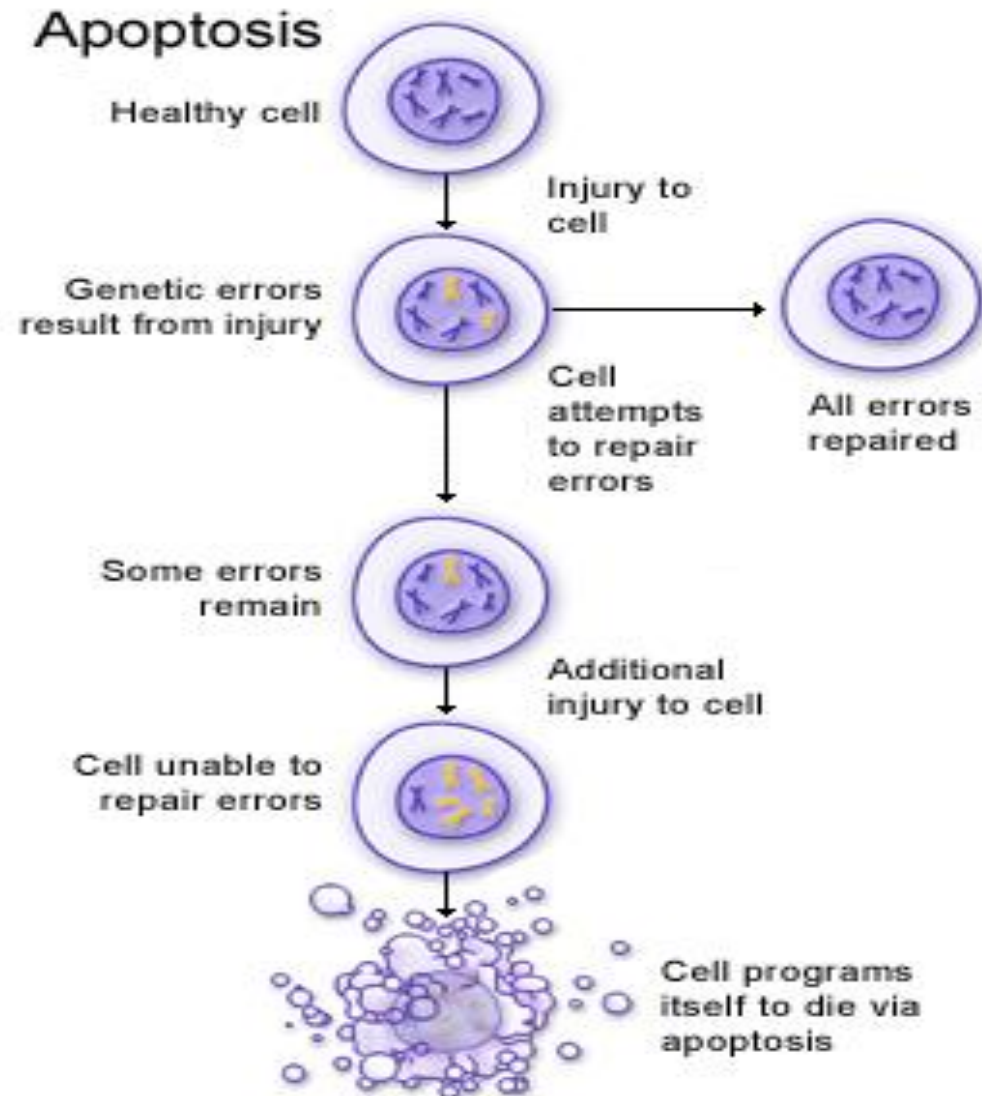


<https://www.biobool.com/news/162.html>

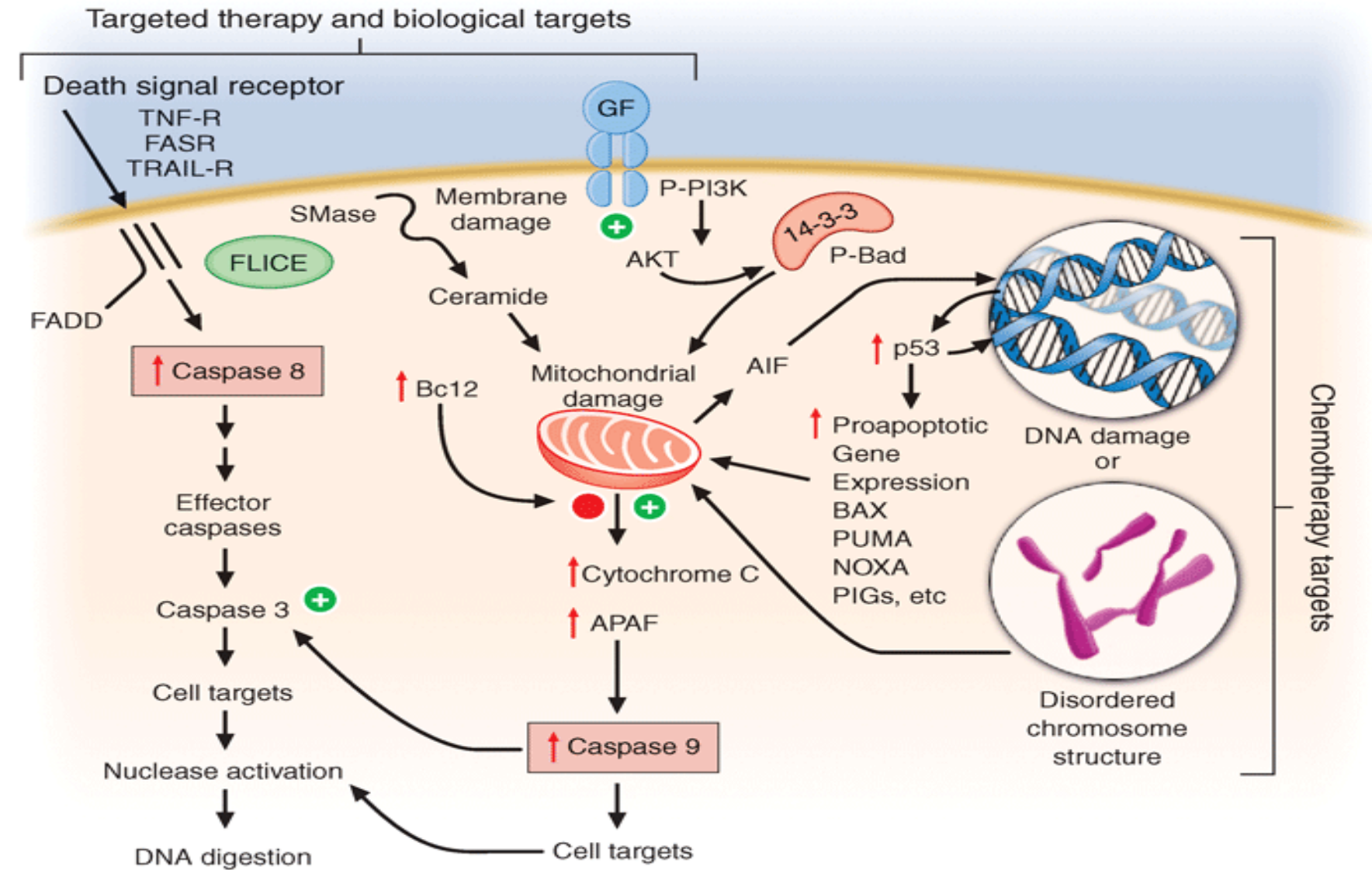
At each cell cycle checkpoint, p53 and other specific proteins monitor for DNA damage, may initiate DNA repair processes, or direct cells down the apoptotic pathway. When cells have normal checkpoint function, drug-induced DNA damage will activate apoptosis at the G1/S or G2/M boundary.

If the p53 gene product or other checkpoint proteins are mutated or absent or the checkpoint function fails, damaged cells will not divert to the apoptotic pathway but will proceed through the S phase and mitosis.

Integration of cell death responses



U.S. National Library of Medicine

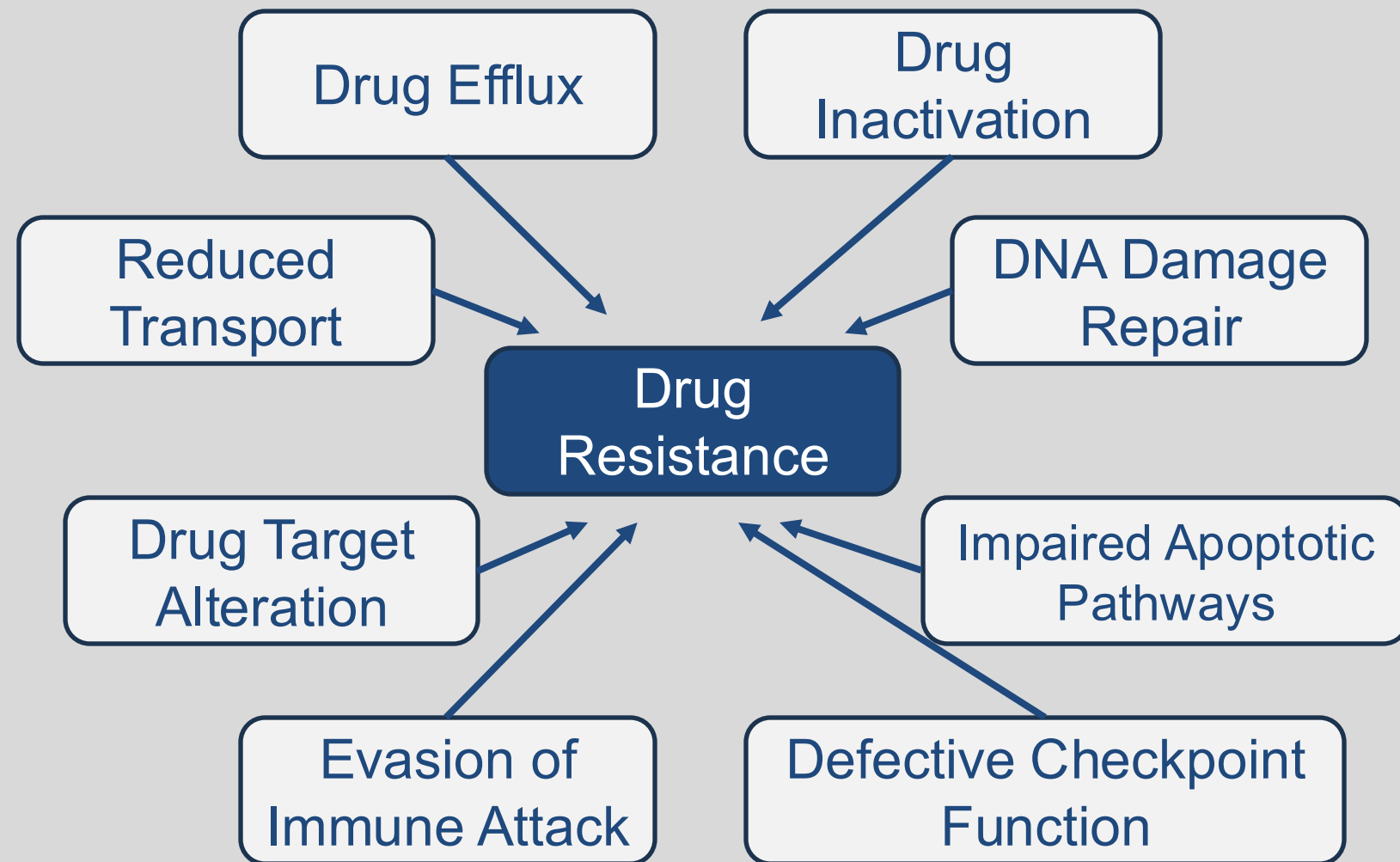


Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine, 18th Edition*: www.accessmedicine.com
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Harrison's Online 18e Figure 85-3

Mechanisms of Resistance

1. Increased rate of DNA repair
2. Defective cell cycle checkpoint function or mismatch repair (MMR) and DNA damage repair systems
3. Impaired apoptotic pathways
4. Decreased drug accumulation:
 - Expression of efflux pumps
 - Reduced transport into cancer cell
 - Inactivation of drugs
 - Decreased activation of prodrugs
5. Increased expression of drug detoxifying proteins, such as glutathione conjugation, that neutralize free radicals
6. Target alterations with decreased affinity for the drugs
7. Tumor evasion of immune attack



Chemotherapy Treatment Strategies

1. Drugs for slow-growing cancers and solid tumors damage DNA.
2. Drugs for rapidly proliferating tumors interfere with DNA synthesis or mitosis.
3. Endocrine therapy for hormone-responsive tumors lowers the amount or interferes with the actions of the hormone, eg, estrogen or testosterone.
4. Multi-drug combinations include cytotoxic drugs acting at specific stages of cell cycle and/or targeted agents that inhibit cancer growth pathways or boost the immune response against the tumor.
 - To maximize the effect and for synergy
 - To minimize the development of resistance
5. Maximum individual doses are administered as frequently as possible, depending on:
 - The time needed for the body to recover from side effects, such as neutropenia
 - The type and stage of the cancer
 - The treatment goals, such as curative or palliative
6. Avoid overlap in major toxicities (ideally)
7. Manage toxicities → antiemetics; analgesics; etc.; non-medical strategies

Toxicities of Antineoplastic Agents

Chemotherapies disrupt DNA function in proliferating normal cells along with tumor cells.

A caution for all treatment regimens:

The pharmacokinetics and toxicities of cancer drugs vary amongst individual patients.

It is *imperative* to:

- recognize toxicities early,
- alter doses or discontinue offending medication to relieve symptoms and reduce risk, and
- provide vigorous supportive care.

Dose-limiting toxicities are specific adverse effects of a drug severe enough to prevent an increase in the dose.

They are critical in determining the maximum tolerated dose during clinical trials, especially in oncology.

Rapidly proliferating cells:

- Bone marrow suppression: ↑ susceptibility to **infection; bleeding**
- GI: Intestinal epithelium: Severe vomiting, mucositis, stomatitis
- Alopecia
- Cutaneous reactions

Tumor lysis syndrome:

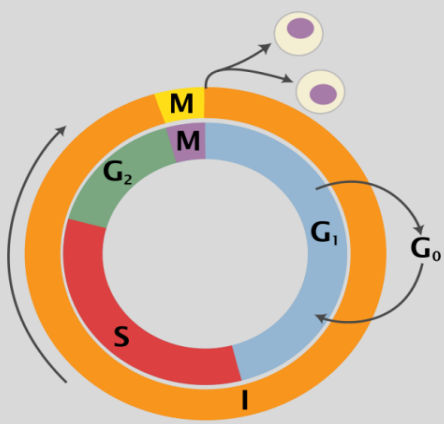
- Metabolic complication of chemotherapy,
- Usually in lymphomas and leukemias

Drug-specific toxicities, acute and delayed

- Cardiac; Pulmonary; Hepatic; Renal; Neuro

Secondary malignancy:

- Leukemia and other malignancies
- Teratogenicity / Fetotoxicity
- Cancer immunotherapy: See Part 2



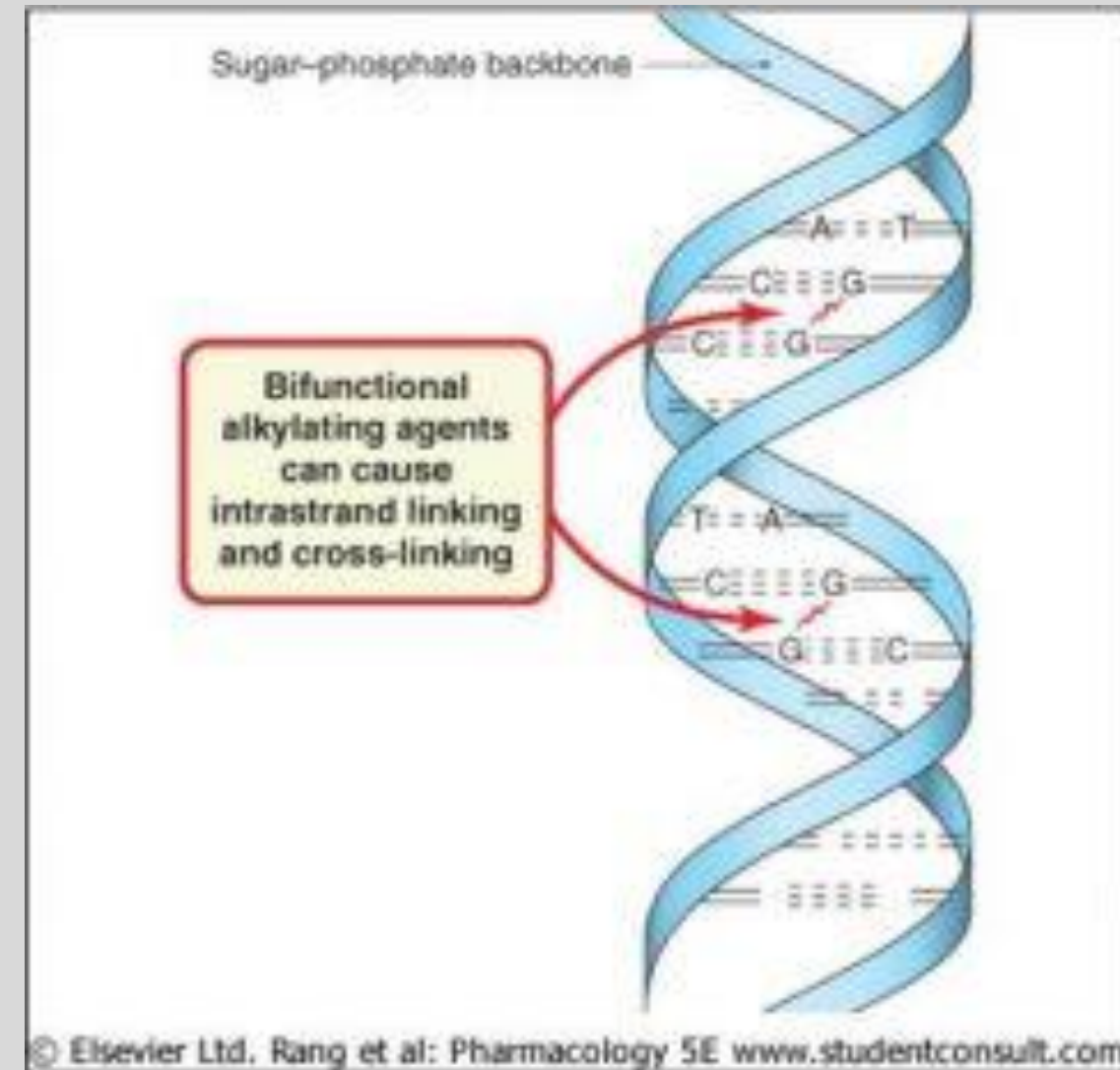
Alkylating Agents

Cell cycle non-specific

There are many drugs in this class.

Alkylating agents interfere with DNA integrity.

Neutrophilic alkylating agents transfer an alkyl ion that attacks an electrophile forming a covalent bond between the alkyl and the electrophile.



Alkylation of DNA: Nucleophilic attack

Cell death depends on cancer cell
recognition of the adduct



**activation of DNA mismatch repair
(MMR) pathway creates DNA
strand breaks by repair enzymes**

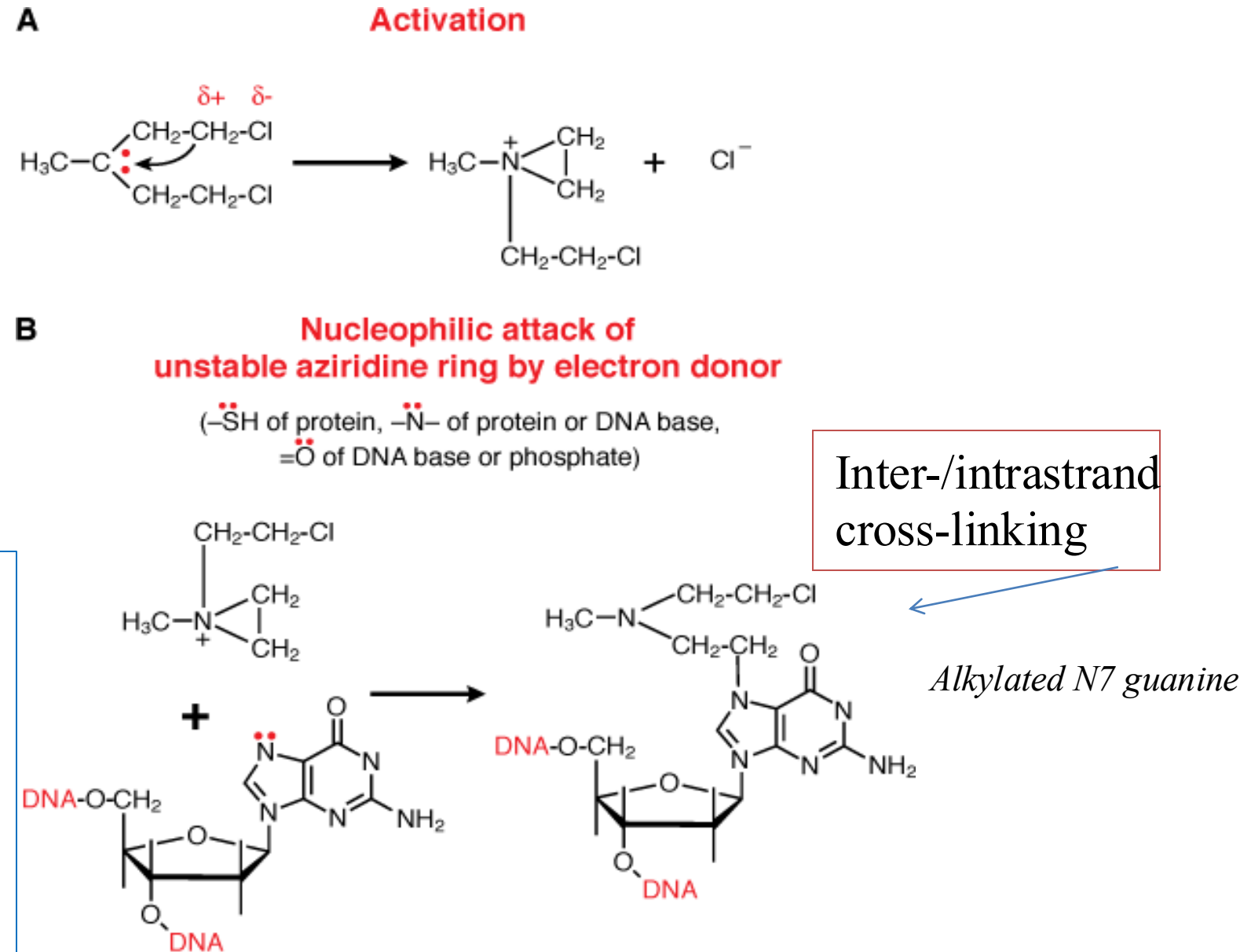


triggers apoptotic response

Essential mechanisms for cell death:

- 1) Recognition of the adduct
- 2) Activation of checkpoint proteins
- 3) Intact MMR / DDR systems, which induce DNA strand breaks by repair enzymes
- 4) Normal genes for p53 and other proteins that trigger apoptosis

**Resistance develops rapidly when a drug is
used as a single agent.**



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*:
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Toxicities of alkylating agents are related to DNA damage of normal rapidly dividing cells and can be quite severe.

Rapidly proliferating cells:

- Bone marrow suppression: DOSE-LIMITING **anemia, infection; bleeding**
- GI mucosal toxicity: **Severe** vomiting, mucositis, stomatitis
- Cutaneous reactions / Hair loss (alopecia)

Tumor lysis syndrome in the treatment of lymphoma and leukemia

- Rapidly lysing cells release contents into blood stream:
- Hyperuricemia, hyperphosphatemia, hyperkalemia, hypocalcemia,
- Acute kidney injury

Organ toxicity

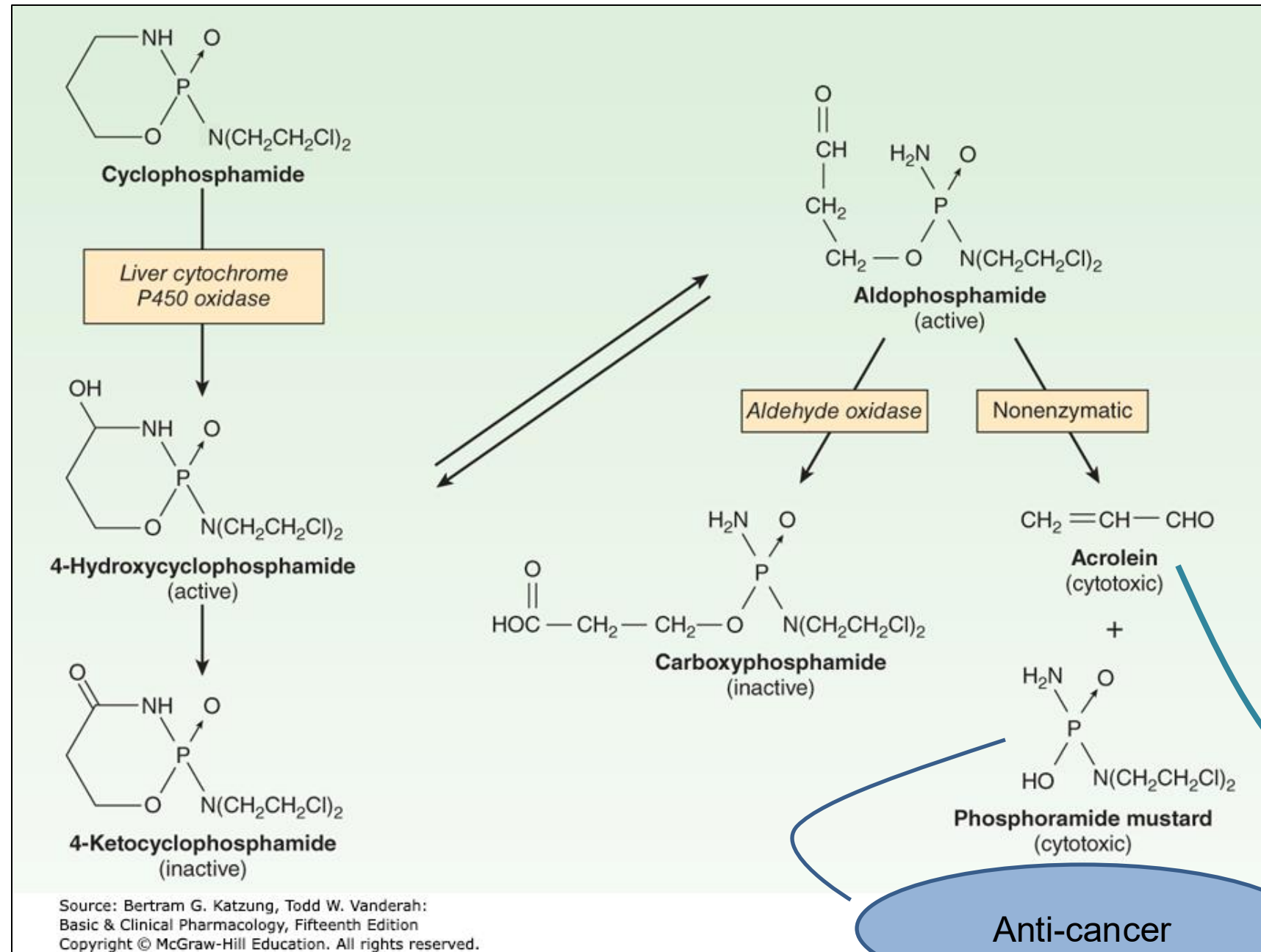
- Hepatic sinusoidal obstruction syndrome
- Pulmonary – “Busulfan lung”
- Peripheral neuropathy
- Extravasation (leakage) from injection site: vesicant (causes blisters / ulcerations)

Secondary malignancy: Leukemia

- **Pregnancy:** Teratogenic
- **Reproductive system:** amenorrhea | azospermia

*Cyclophosphamide
Ifosfamide

Pharmacokinetics: prodrug → cytotoxic phosphoramidate mustard & urotoxic metabolite



- I.V.; oral (high bioavailability)
- Prodrug
- Metabolic activation via CYP2B6 and other CYPs

hydroxycyclophosphamide
↓
aldophosphamide
↓
phosphoramidate mustard
(active chemotherapy)
+
Acrolein
(toxic to bladder epithelium)
↓
Hemorrhagic
cystitis

Anti-cancer

Cyclophosphamide / Ifosfamide Mechanisms

Cytotoxicity

Alkylation of nucleophilic groups on DNA bases, esp the N7 position of guanine →

- Inter-, intrastrand cross-linking,
- abnormal base pairing,
- miscoding,
- strand breakage, or
- depurination by excision of guanine residues

Resistance

Increased expression of **Aldehyde dehydrogenase** → increased rates of metabolic degradation to inactive metabolites

Increased expression of **Glutathione transferases** → detoxification of most alkylating intermediates

Therapeutic Uses

The TAKEAWAY:

Cyclophosphamide is widely used for treatment of solid tumors, hematologic cancers and it is a potent immunosuppressive, making it useful in autoimmune disorders. Ifosfamide has more limited uses in the treatment of some solid tumors and lymphomas.

Cyclophosphamide

- **Broad use** in adults and pediatrics for a variety of
 - hematologic cancers and
 - solid tumors
- **Potent immunosuppressive** properties
⇒ treatment of autoimmune disorders, e.g.
 - Wegener's granulomatosis,
 - Rheumatoid arthritis, and
 - Nephritic syndrome

Ifosfamide

- Testicular cancer
- Lymphomas
- Bone, tissue sarcomas
- Bladder
- Cervical
- Ovarian
- Lung cancer

Cyclophosphamide drug-specific serious toxicities

In addition to alkylating agents class effects (slide 22)...

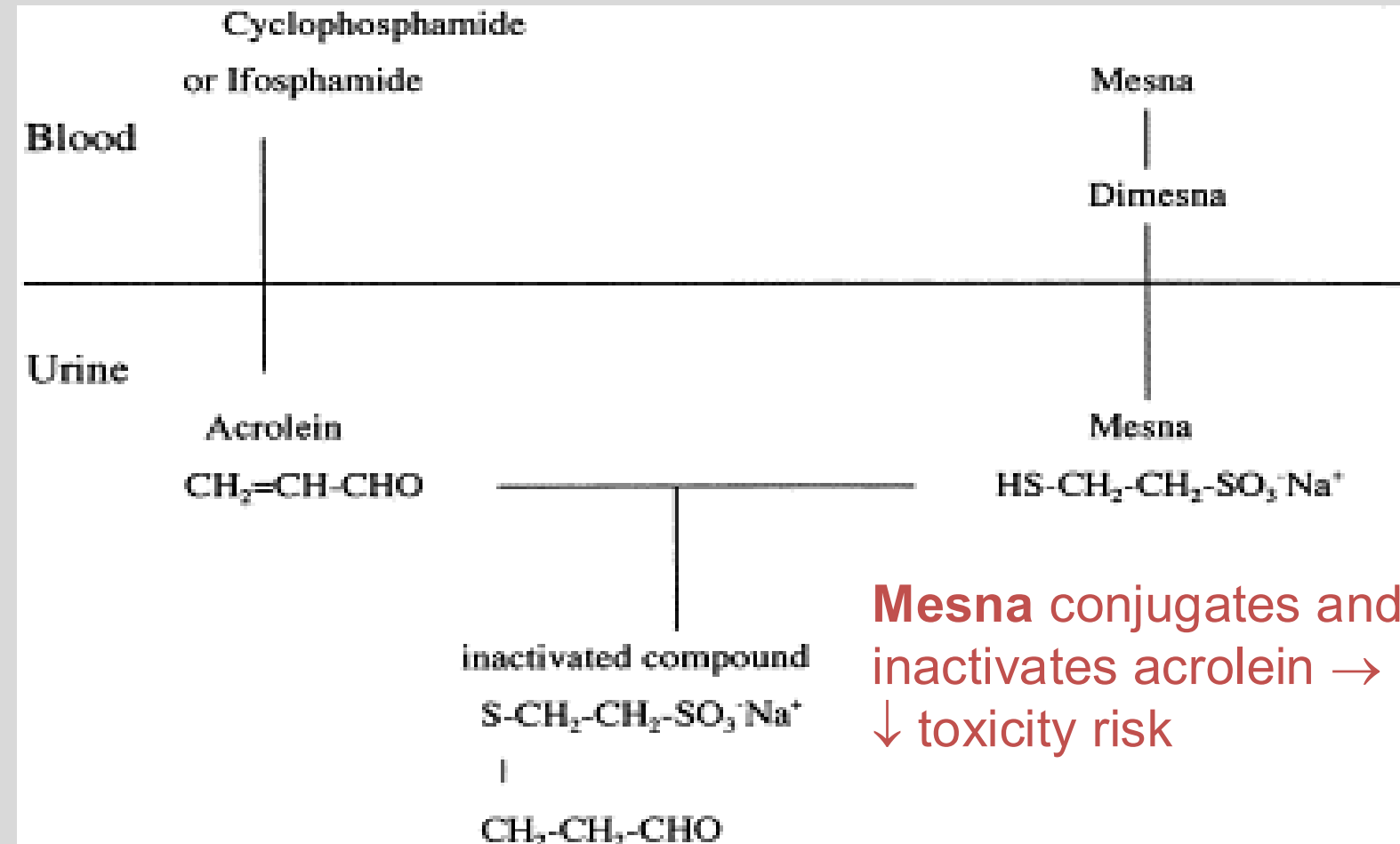
- Hemorrhagic cystitis due to acrolein toxicity to bladder epithelium

Mitigation: Hydration + Mesna

- Bone marrow suppression
- Congestive heart failure

May exacerbate effects of other cardiotoxic medications and underlying myocardial dysfunction.

- Arrhythmias



Check your knowledge

What class of drugs induces cell death by nucleophilic attack on DNA bases, esp. the N7 guanine, leading to inter- and intrastrand cross-linking, abnormal base pairing, miscoding, and strand breakage.

What cellular mechanisms are essential for cell death?

What is the active cytotoxic metabolite of cyclophosphamide (and ifosfamide)?

Check your knowledge

What is the name of the urotoxic metabolite and how can the toxicity be mitigated?

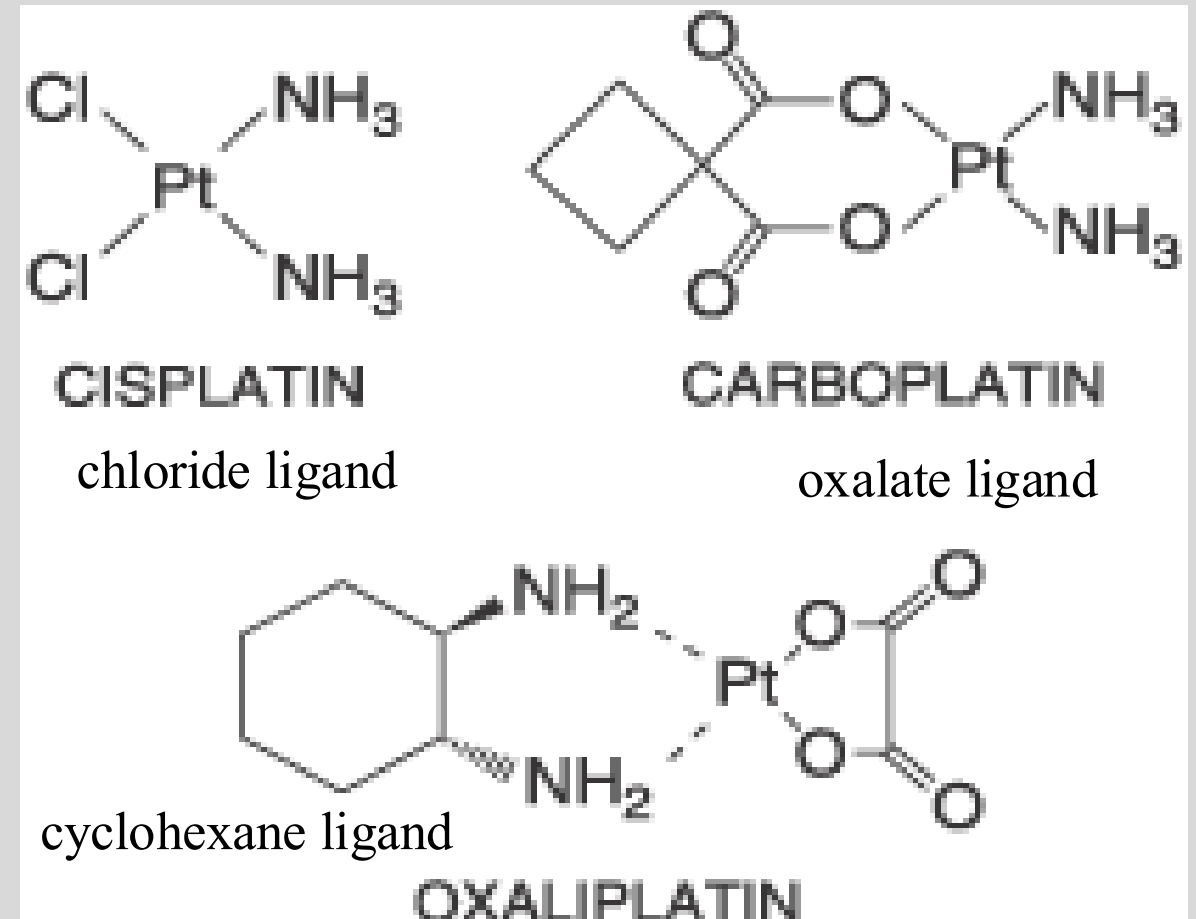
What is a mechanisms of tumor resistance to cyclophosphamide and ifosphamide?

Cyclophosphamide is used in the treatment of what categories of cancers?

List the common and serious toxicities of cyclophosphamide.

Platinum Coordination Complexes

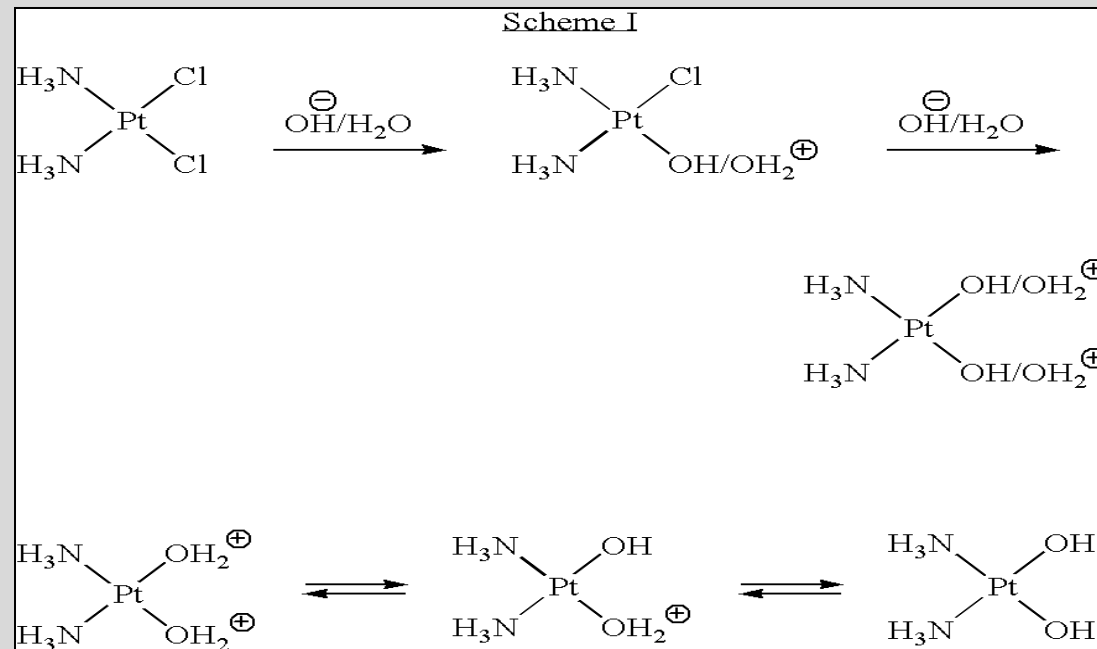
- *Cisplatin
- *Carboplatin
- *Oxaliplatin



1. Passive diffusion and Cu^+ active transporter $\rightarrow$
2. Aquation (activation) $\rightarrow$
3. Reacts with purine – mainly N7 guanine – either on the same or neighboring strands of DNA $\rightarrow$

- replication/transcription inhibition
- cell cycle arrest
- strand breaks
- apoptosis – if p53 and other checkpoint proteins are intact

Platinum analogs also bind nuclear and cytoplasmic proteins.



Above:

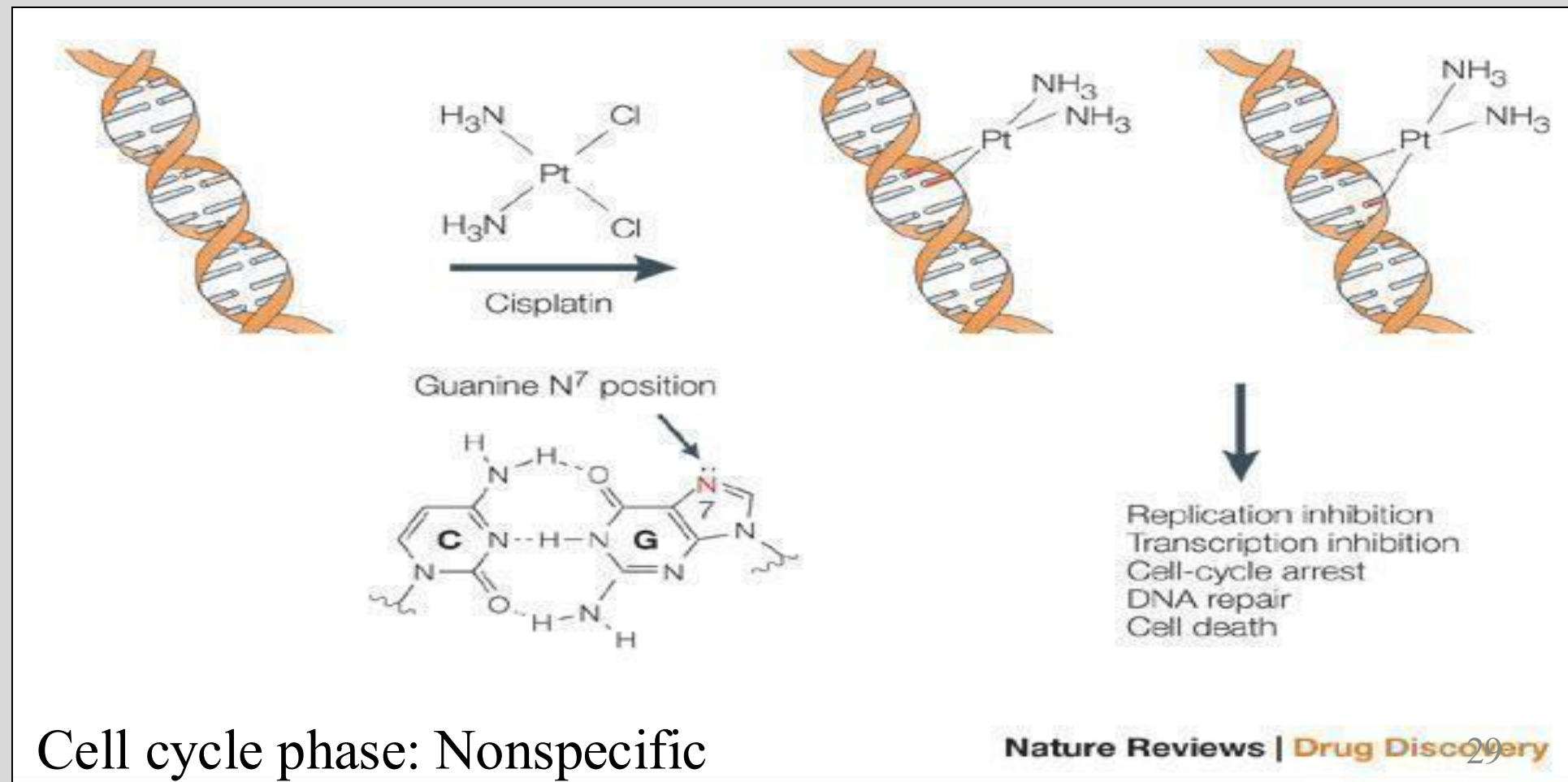
<http://www.google.com/patents/US7759488>

Below:

Nature Reviews Drug Discovery 4, 307-320

(April 2005) doi:10.1038/nrd1691

http://www.nature.com/nrd/journal/v4/n4/fig_tab/nrd1691_F3.html



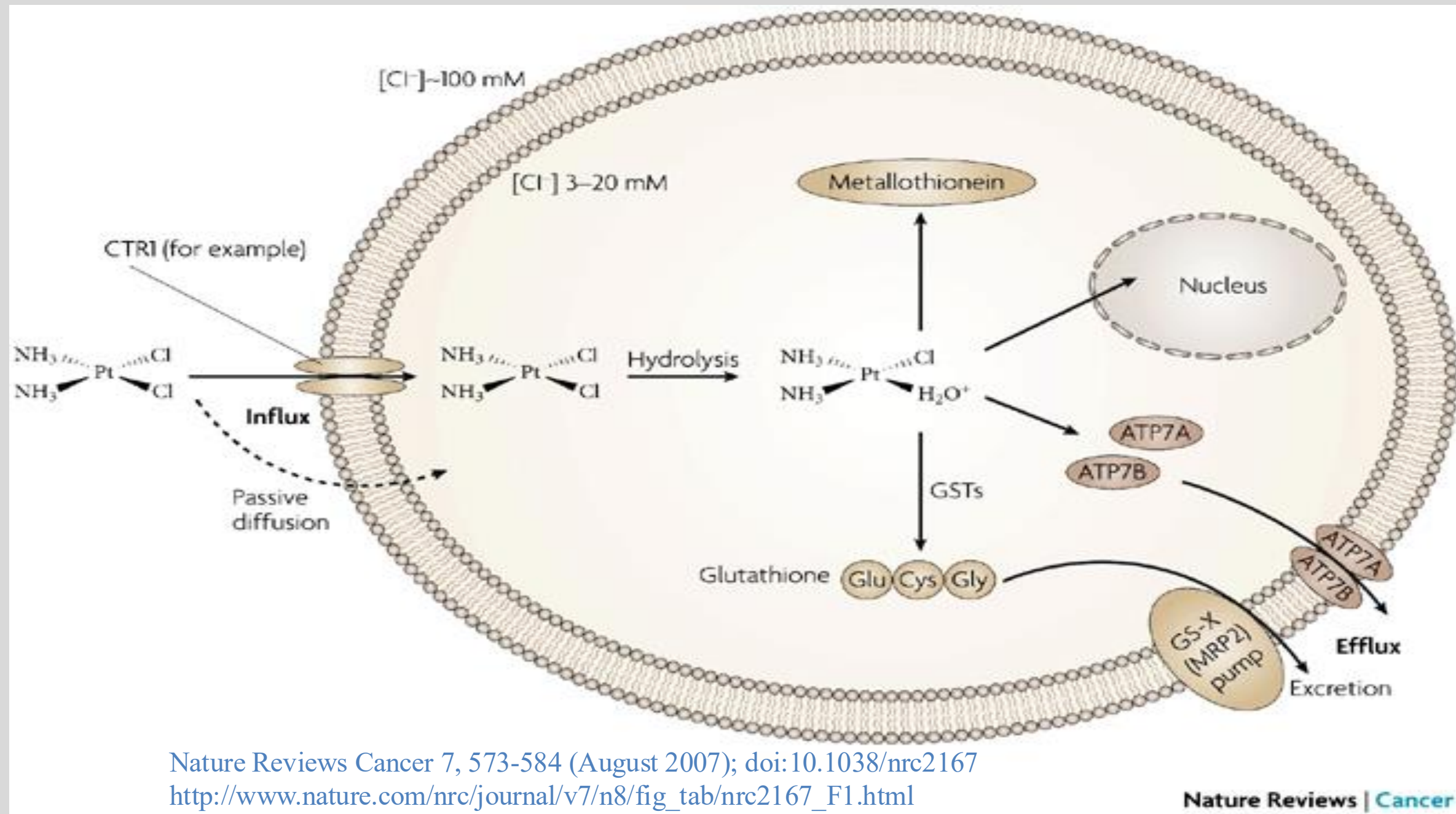
Resistance to Platinum Analogs

Decreased uptake	Efflux	Inactivation	Repair
CTR1 copper transporter 1. Down-regulation, and 2. ↓ expression Decreased uptake is the predominant mechanism of resistance	ATP7A and ATP7B copper transporters Physiologic function: removal of excess copper from the cell	↑ intracellular levels of 1. Glutathione, 2. Metallothionein, 3. Other sulfhydryls These molecules bind to and inactivate the drugs.	Overexpression of: 1. Nucleotide excision repair components, 2. Base-excision repair, 3. Mismatch repair (MMR), 4. double-strand-repair breaks

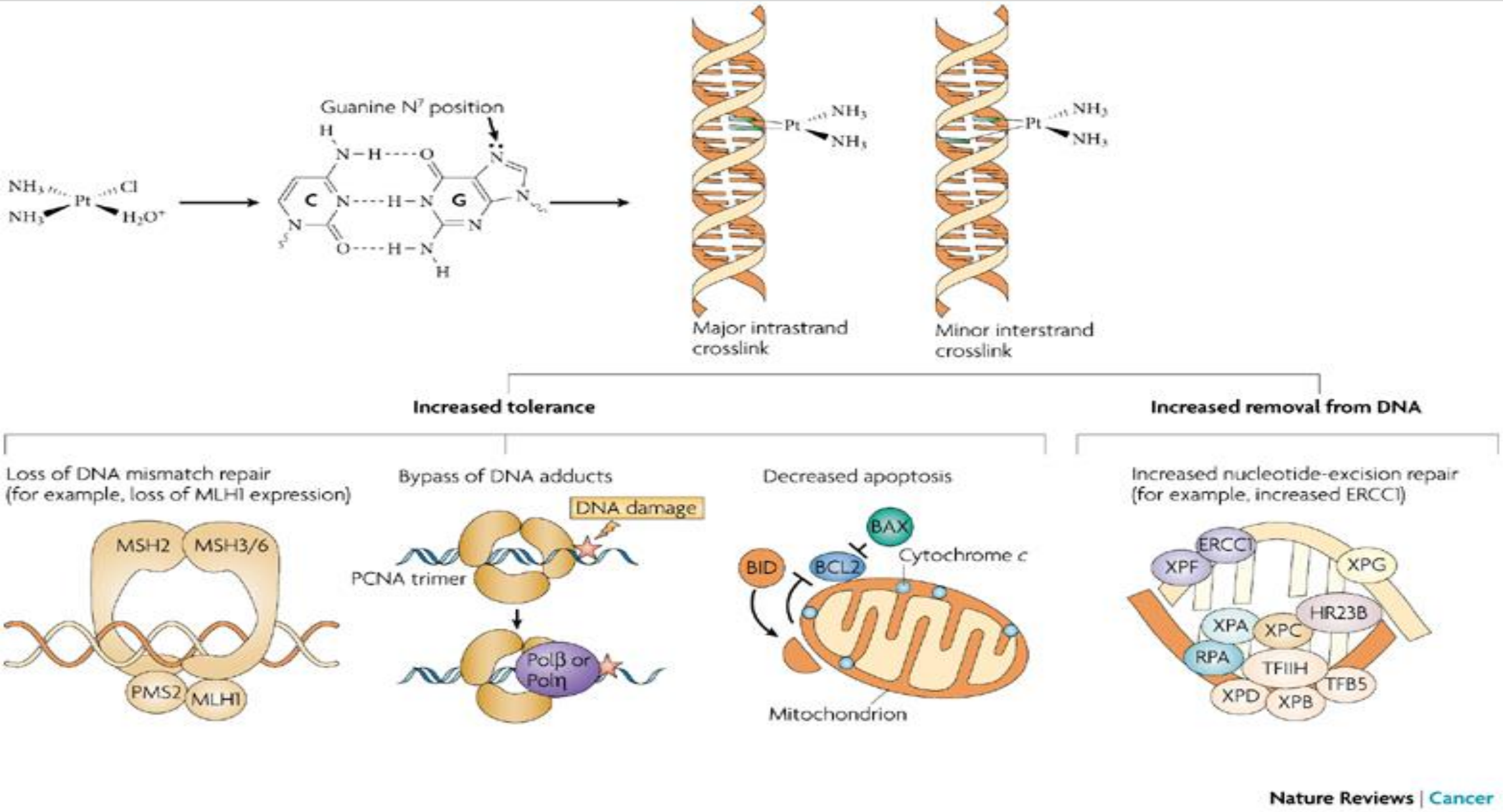
Cross-resistance may occur between cisplatin and carboplatin
Tumors resistant by MMR are not cross-resistant to oxaliplatin

Platinum Analogs Mechanisms of Resistance

↓ Influx, Inactivation, and ↑ Efflux



Platinum Analogs Mechanisms of Resistance After DNA Binding



Platinum Analogs

Common features	IV, poor CNS penetration renal clearance dosage adjustment in renal impairment	Synergy with radiation therapy Interaction: T before P Administer taxane before platinum to limit myelosuppression and increase efficacy
Cisplatin	Covalent binding to plasma proteins Nonenzymatic inactivation (unbound fraction) Vigorous saline hydration $t_{1/2\text{terminal}}$ up to 5 days	Major antitumor activity in a broad range of solid tumors
Carboplatin	Less reactive Less renal and GI toxicity and vigorous IV hydration is not required (but recommended)	Broad-spectrum activity against a wide range of solid tumors
Oxaliplatin	Highly reactive Cytotoxicity does not depend on active mismatch repair system for efficacy. (Bulky ligands create a different distortion pattern of DNA helix.)	Colorectal and gastric cancer Oxaliplatin is a component of first-line chemotherapy of advanced colorectal cancer (FOLFOX)

Dose-limiting toxicities (in red) and other major toxicities

Reminder: Being aware of major toxicities is important for optimizing the patient's therapy. Patients should be monitored for them, and the toxicities should be mitigated, if possible.

Cisplatin

Highly emetogenic – Vomiting: persistent, acute and delayed neurotoxicity; ototoxicity; myelosuppression, peripheral neuropathy

Nephrotoxicity vigorous chloride diuresis with normal saline + diuretic to reduce toxicity

Carboplatin

More tolerable – Toxicities of cisplatin but less severe.

Myelosuppression

Oxaliplatin

Hypersensitivity / Anaphylaxis – can occur within minutes
Toxicities similar to cisplatin but less severe

Peripheral neuropathy acute (exacerbated by cold) and delayed progressive neuropathy

All: Leukemia, pulmonary fibrosis months to years later

Check your knowledge

How does the mechanism of the platinum analogs compare with the mechanism of the alkylating agents?

What are the main mechanisms of resistance to platinum analogs?

Check your knowledge

What is the definition of dose-limiting toxicities?

What are the dose-limiting toxicities of cisplatin, carboplatin, and oxaliplatin?

In sequential administration of a platinum analog with paclitaxel or docetaxel, why is the taxane administered first?

Antimetabolites

S phase of cell cycle

Definition: An antimetabolite is a substance that is similar in structure to a metabolite or enzymatic substrate, but interferes with the normal division and function of cells

- Antimetabolites are analogs and competitive antagonists of the natural substrates – folic acid, pyrimidines, purines.
- Antimetabolites inhibit nucleotide synthesis, which depletes the pool of nucleotides necessary for RNA and DNA synthesis.
- Antimetabolites induce cell death in rapidly dividing cells by interfering with DNA synthesis (S phase of the cell cycle).

Antimetabolites Class Toxicities – Acute and Delayed

Like the alkylating agents, antimetabolites can cause:

- Bone marrow suppression – anemia, neutropenia, thrombocytopenia
- GI toxicity – nausea, vomiting, diarrhea, mucositis, stomatitis
- Hepatotoxicity – ranging from elevated serum transaminases to liver failure
- Pulmonary toxicity – pneumonitis, interstitial fibrosis
- Nephrotoxicity – impaired kidney function and interstitial toxicity as drug deposits in the kidney parenchyma and tubular epithelial cells
- Neurotoxicity – peripheral neuropathy
- Cutaneous reactions – rash, hand-foot syndrome, alopecia

Folate Antagonists

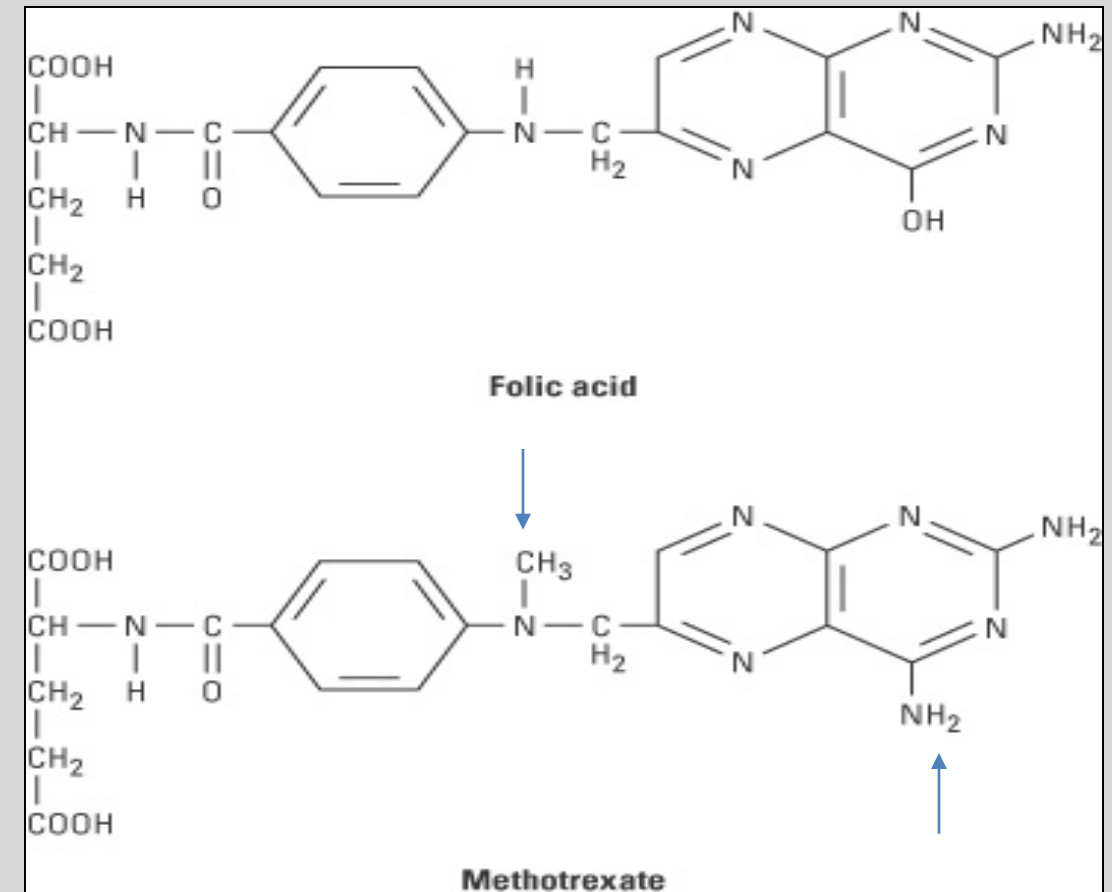
*Methotrexate

Pemetrexed

Pralatrexate

Folic acid is an essential dietary factor (vitamin B9) converted by enzymatic reduction to FH_4 cofactors that provide methyl groups for the synthesis of thymidylate and purines, precursors of RNA and DNA.

Folic acid is essential for DNA synthesis and repair, amino acid metabolism, red blood cell formation, and neural tube development in the fetus.



Methotrexate (MTX) Pharmacokinetic Properties

	Property	Comment
A	IV administration for cancer treatment; intrathecal (IT) for ALL	Oral, IM, subQ for weekly nononcologic immunosuppressant indications
D	Taken up by reduced folate transporter ~50% plasma protein bound Narrow therapeutic window Distributes in body water Poor CNS penetration Sustained concentrations in liver/kidney	Drugs that displace methotrexate from albumin → ↑ risk of severe myelosuppression MTX is a Substrate of BCRP, P-gp, OATP (liver) BCRP: breast cancer resistance protein, an efflux transporter expressed in brain, GI tract, placenta, and testes and eliminates substrates from liver via biliary tract and via transport into the proximal tubule.
M	Intracellular polyglutamation Low intestinal and hepatic metabolism	Active MTX-PG accumulates intracellularly and is slowly eliminated (ion trapping inside cell)
E	Urine, 80%-90% unchanged drug	Dose adjustments for renal impairment
t _{1/2}	Age and dose-dependent	Children: 1-6 h (ALL treatment) Adults: 8-15 h (cancer treatment)

MTX and MTX-PG Mechanisms

Cell cycle: S phase

1. MTX / MTX-PG block DHFR → ↓ FH₄ synthesis.
↓ FH₄ levels reduces synthesis of thymidylate and purine nucleotides necessary of RNA and DNA synthesis.
2. MTX / MTX-PGs levels increase behind blocked DHFR
→ Inhibition of additional targets including TS, GART, AICAR transformylase

↓
Cell cycle arrest, activation of p53, apoptosis

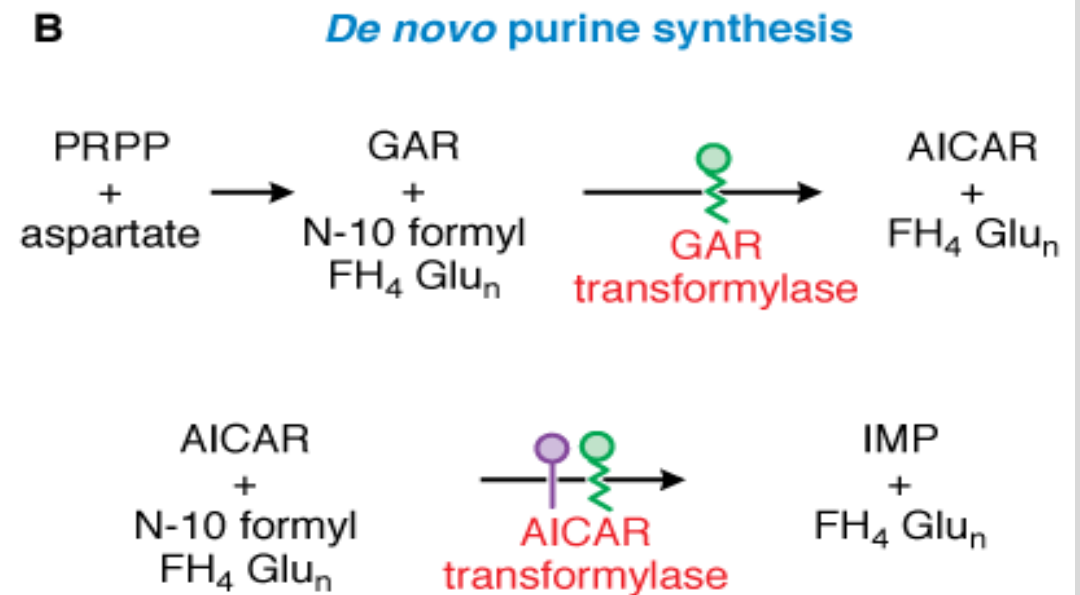
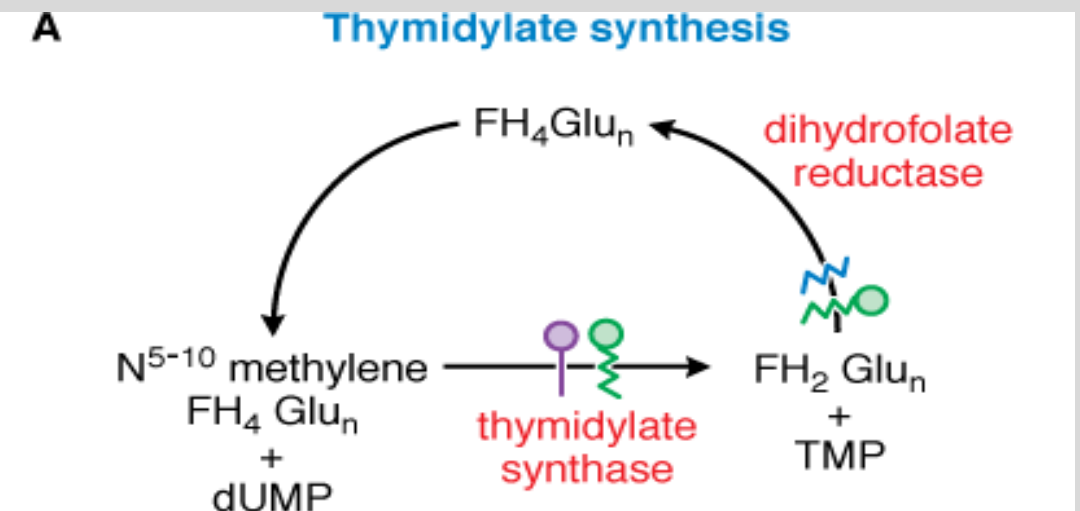
↓
Cell death in S phase

DHFR: dihydrofolate reductase catalyzes conversion of folic acid to FH₄

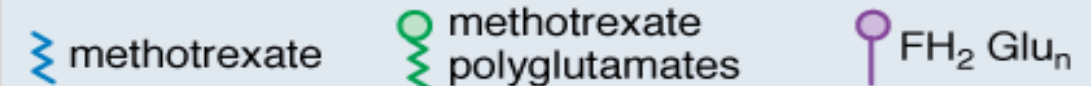
TS: thymidylate synthase, synthesis of thymidylate (thymidine monophosphate, TMP)

GART: glycinamide ribonucleotide transformylase, intermediate in purine synthesis

AICAR: aminoimidazole carboxamide ribonucleotide, intermediate in IMP synthesis



REACTION INHIBITED BY:



Source: Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition: www.accessmedicine.com

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MTX Resistance

Impaired transport of drug into the cells

Folic acid and analogs are polar. They are taken into cells by transport systems.
MTX, ralitrexed, and pemetrexed are taken up by reduced folate transporters.

↑ Expression of drug efflux transporter of the MRP class

MRP=multidrug resistance-associated protein-1

Production of altered DHFR with reduced affinity for methotrexate

↑ Concentrations of DHFR through gene amplification or altered gene regulation

↓ Ability to synthesize methotrexate polyglutamates

Methotrexate has many therapeutic uses.

Oncology uses: **Broad spectrum:** solid and liquid tumors, including childhood ALL

Autoimmune disease: Weekly low dose for immunosuppressive effects

Abortifacient: Tubular ectopic pregnancy

Methotrexate drug-specific toxicities, in addition to class toxicities

Bone marrow suppression

Mucositis, stomatitis, nausea, vomiting, diarrhea

Hepatotoxicity is cumulative and progressive

Pneumonitis can occur at any time

Neurotoxicity with high IV dose:

- paralysis, palsy, seizure, coma
- arachnoiditis with intrathecal admin
- demyelinating encephalitis with radiation

Pregnancy: Fetal loss - abortifacient

Reproductive toxicities

Impaired fertility- defective oogenesis / spermatogenesis

Leucovorin rescue: Converted to active methylene-FH₄ → “rescue” host cells from:
folinic acid Bone marrow toxicity, hepatotoxicity, mucositis/stomatitis

Leucovorin does not reverse neurotoxicity or pulmonary toxicity.

Pyrimidine analogs

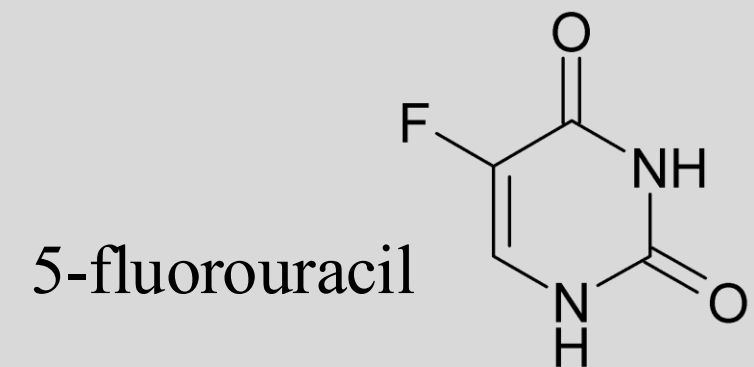
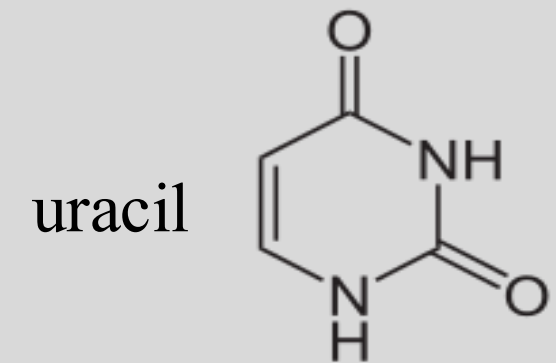
Fluoropyrimidine

Antimetabolites

*5-Fluorouracil (5-FU)

*Capecitabine (oral prodrug of 5-FU)

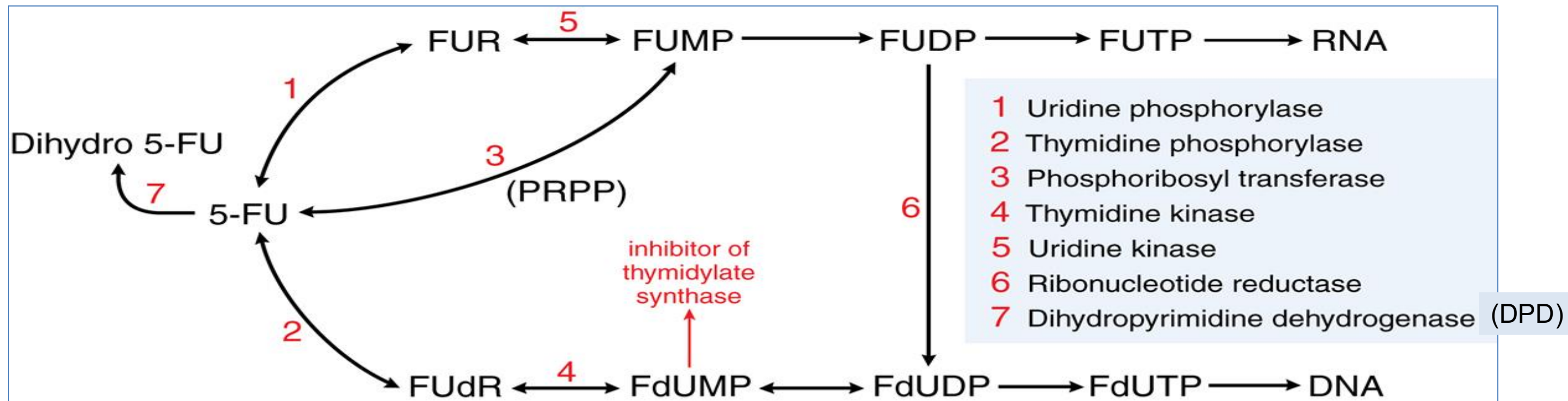
Floxuridine (FUdR)



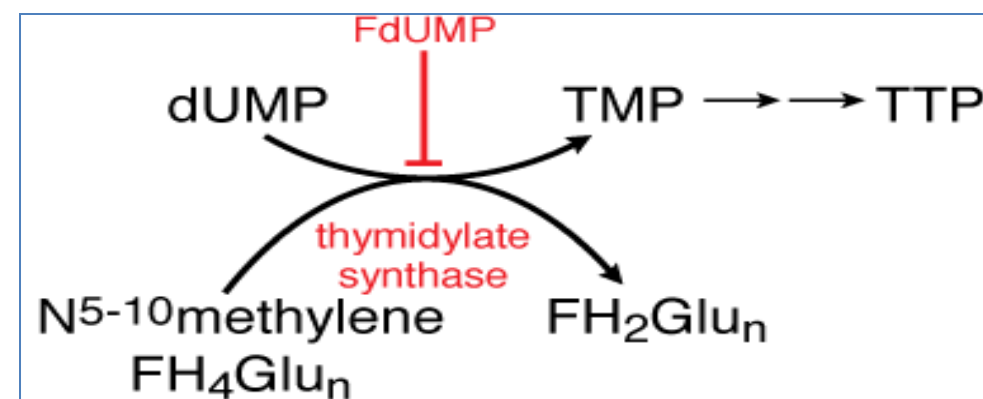
5-FLUOROURACIL

Activation	Enzymatic ribosylation and phosphorylation of 5-FU to the active nucleotides F-UMP and F-dUMP
Distribution	Throughout body, including CSF and brain tissue
Metabolism	5-FU dihydropyrimidine dehydrogenase (DPD) → 5-6-dihydrofluorouracil DPD is found in liver, intestinal mucosa, tumor and other tissues.
Excretion	Partially metabolized to CO ₂ , which is excreted by the lung Urine, 5% unchanged drug
t _{1/2} plasma	6-20 minutes → Administered by continuous IV infusion Biologic t _{1/2} : prolonged, tissue dependent
CAPECITABINE	Oral prodrug of 5-FU; converted primarily in the liver to inactive intermediate compounds, then to 5-FU in the tumor
Pharmacogenomics	Patients with a complete deficiency of DPD (autosomal recessive trait) are at risk of severe, excessive toxicity. This enzyme deficiency is seen in up to 5% of cancer patients.

5-FU Mechanisms of Action – “Thymine-less death”



1. 5-FU → → FUMP → FUDP ribonucleotide reductase → FdUDP ↔ FdUMP → FdUMP-folate-TS 3-way complex → inhibits TS activity; thymidylate synthesis is inhibited
2. FdUDP → FdUTP → incorporated into DNA → DNA strand breaks
3. 5-FU → FUMP → FUDP → FUTP → incorporated into RNA → RNA processing and function is disrupted
—5-FU cytotoxicity is thought to be mediated by the combined effects of both DNA- and RNA-mediated events.—



Other actions of 5-FU nucleotides:

- Inhibition of RNA processing
- Incorporation into DNA

Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics*, 12th Edition: www.accessmedicine.com
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Resistance | Therapeutic uses | Toxicities

Resistance	Therapeutic Uses Broad spectrum, solid tumors	Toxicities
Decreased activation - Cancer cells that do not convert 5-FU to 5-FdUMP Altered substrate - Altered or increased thymidylate synthase levels	First line regimen: Metastatic colorectal cancer Synergy: FOLFOX or FOLFIRI Folinic acid—5-FU—thymidylate form ternary complex Topical 5-FU: Actinic keratosis Basal cell carcinoma, superficial Squamous cell carcinoma, in situ Penile carcinoma, in situ Fol: Folinic acid (leucovorin) F: 5-FU OX: Oxaliplatin IRI: Irinotecan	Class toxicities: - Bone marrow suppression, GI, <i>CNS neurotoxicity</i> Hand-foot syndrome - especially with capecitabine - Palmar-plantar erythema: desquamation, pain, and sensitivity to touch Coronary artery vasospasm leading to cardiac ischemia - Angina, MI, arrhythmia, heart failure - Rare, can be fatal.



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Stomatitis induced by chemotherapy
(painful erosions and ulcerations)



Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine*, Second Edition: www.accessmedicine.com
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Palmar-Plantar Erythema / Desquamation



[Wikipedia: Chemotherapy-induced acral erythema](https://en.wikipedia.org/wiki/Chemotherapy-induced_acral_erythema)

Pyrimidine Analogs

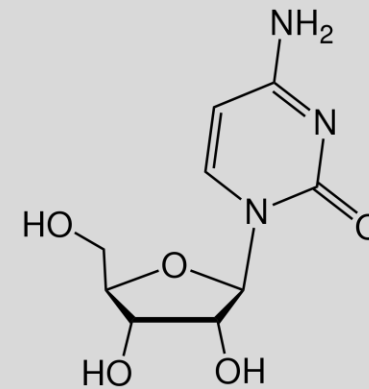
Cytidine Analogs

*Cytarabine (Ara-C), an aribinoside

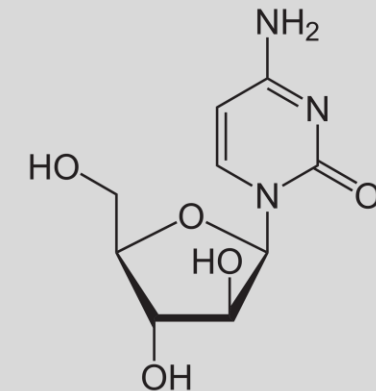
Gemcitabine (difluouro analog of deoxycytidine)

Azacitidine

Decitabine

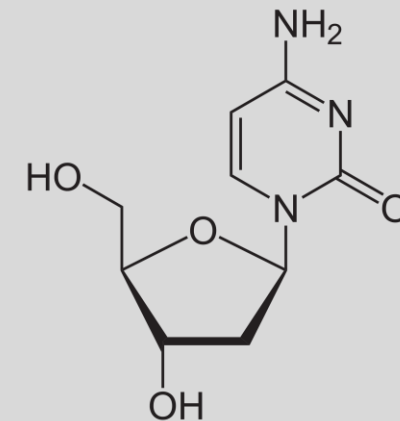


Cytidine

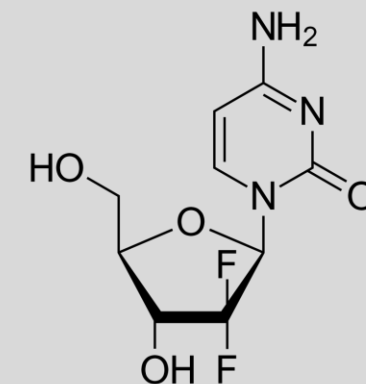


Cytarabine

arabinose ring instead
of a ribose ring



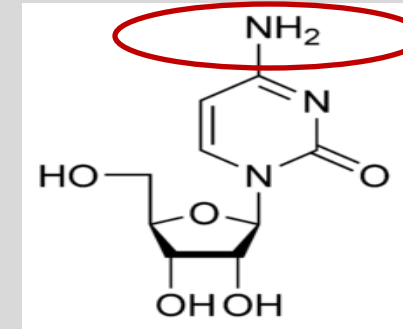
Deoxycytidine



Gemcitabine

Cytarabine / Gemcitabine Class Pharmacokinetics

- **Parenteral**
- Distributes to CNS
- Extensive degradation by **cytidine deaminase** to inactive uracil metabolites
 - Primarily in liver
 - Cytidine deaminase activity is high in many normal tissues
- Urine as inactive uracil metabolites
- $t_{1/2}$ dependent on dose and duration



🔑 Efficacy: Cytidine deaminase activity is **lower in AML cells** and other human tumors.

🔑 Due to the formation of intracellular nucleotides, which become incorporated into DNA, the **effects persist** for many hours.

Cytarabine Mechanisms of Action and Resistance

Mechanisms of Action:

Enters cells via nucleoside transporter →

Cytarabine **deoxycytidine kinase** → Ara-CMP → Ara-CDP → Ara-CTP

1. competitive inhibition of:
 - A. DNA polymerase- α (blocks DNA synthesis), and
 - B. DNA polymerase- β (prevents DNA repair)
2. incorporation into DNA → blocks chain elongation → activation of checkpoint kinases → apoptosis

Mechanisms of Resistance

- ↓ **Conversion to active form:** Loss of deoxycytidine kinase (primary mechanism)
- ↓ **Transport:** Reduced intracellular levels of cytosine analog
- ↑ **Deactivation:** Increased concentration of deactivation enzymes, cytidine deaminase and others may be involved

Cytarabine Therapeutic Uses and Toxicities

Although structurally related, deoxycytidine analogs have different spectrums of action.

Cytarabine

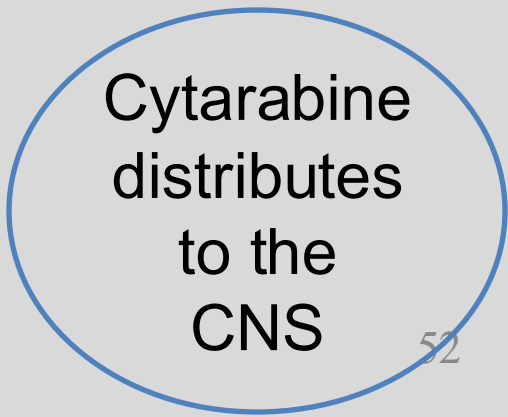
AML: *drug of choice* with daunorubicin or idarubicin

Other leukemias and high-grade lymphomas

Cytarabine has no activity in solid tumors

Toxicities (typical of antimetabolites class)

- Myelosuppression: acute severe leukopenia, thrombocytopenia, and anemia with megaloblastic changes
- Dyspnea, pulmonary infiltrates; can be fatal
- Mucositis; nausea/vomiting; rash; alopecia
- Cerebellar toxicity, high dose or intrathecal (ataxia; slurred speech)
- Cerebral toxicity, high dose or intrathecal (seizures, coma)



Cytarabine
distributes
to the
CNS

Purine Analogs

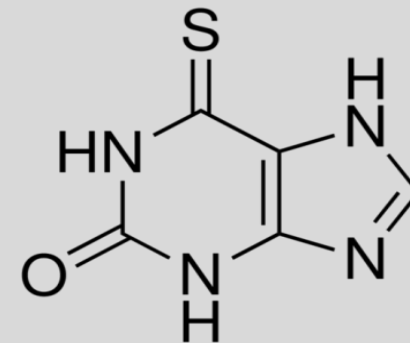
* 6-Mercaptopurine (6-MP)

6-Thioguanine (6-TG)

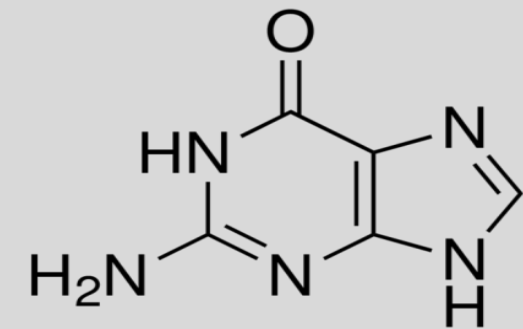
Fludarabine

Cladribine

Pentostatin

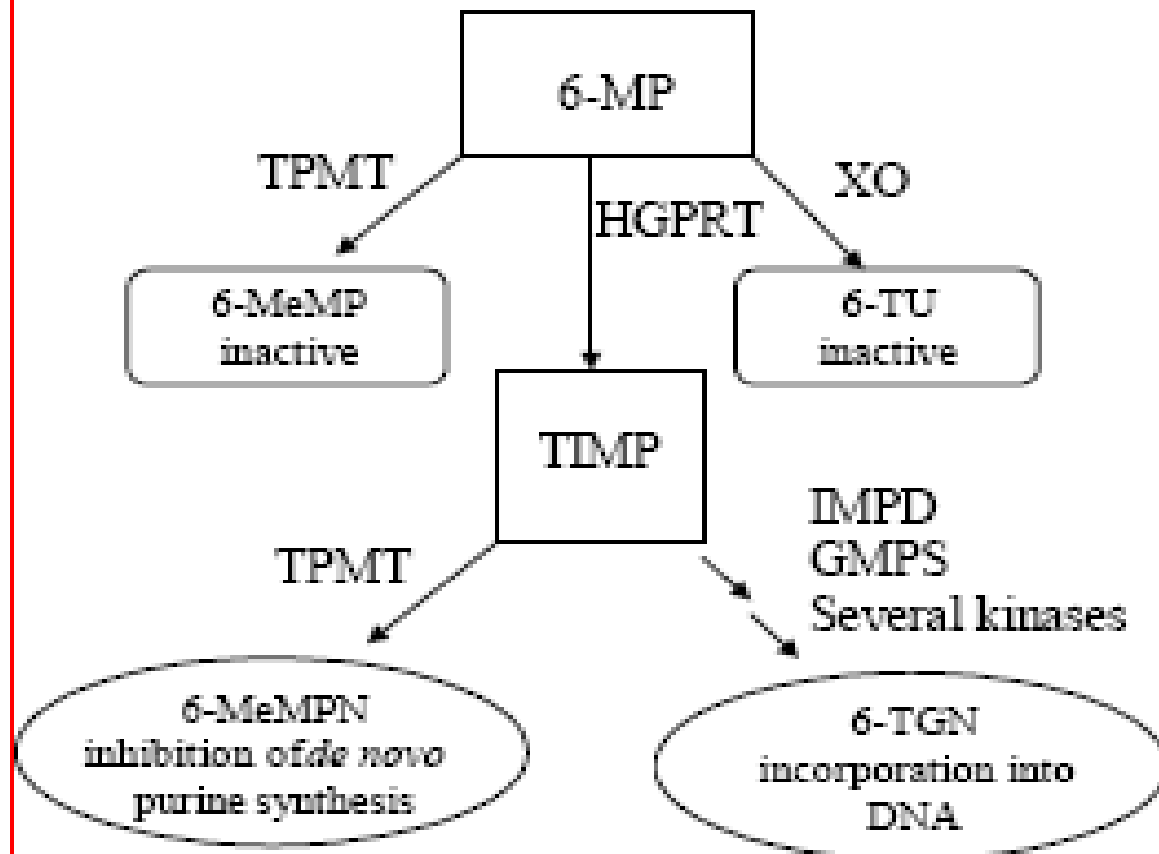


6-mercaptopurine



guanine

Cell cycle: S phase



TPMT: thiopurine methyltransferase

XO: xanthine oxidase

TIMP: thioinosine monophosphate

IMPD: inosine-5'-monophosphate dehydrogenase

GMPS: guanosine monophosphate synthase

6-TGN: 6-thioguanine nucleotides

6-MeMPN: 6-methylmercaptopurine nucleotides

- 6-MP hypoxanthine-guanine phosphoribosyl transferase (HGPRT) → 6-thioinosinic acid → 6-TIMP → → 6-TG nucleotides →

1. TIMP inhibits

- several enzymes of *de novo* purine nucleotide biosynthesis, and
- conversion of IMP to purine nucleotides

2. 6-TGN incorporation into DNA → strand breaks and base mispairing

Pharmacogenetic variability:

- TPMT slow metabolizers

Drug interaction:

- Xanthine oxidase inhibitors



Increased risk of 6-MP toxicity
(potentially severe)

Mercaptopurine

Therapeutic Uses

- Acute lymphoblastic leukemia (ALL)
- Component of various regimens

Immunosuppressant

Potent T cell suppression

- Crohn's disease

Toxicities

- Bone marrow suppression
- Hepatotoxicity
- GI symptoms
- Risk of infection

Drug interaction

Allopurinol inhibits xanthine oxidase → shunts 6-MP to formation of more toxic metabolite

🔔 Decrease 6-MP dose to ~25% of usual dose

Check your knowledge

What structural and mechanistic features are common to the antimetabolites class?

How does inhibition of dihydrofolate reductase lead to cell cycle arrest?

Check your knowledge

How does leucovorin reduce bone marrow, liver, and GI toxicity of methotrexate therapy?

What is the mechanism of action for fluorouracil?

Check your knowledge

What is the function of leucovorin in the FOLFOX regimen for colorectal cancer treatment?

What drug is a cytidine analog activated to the triphosphate deoxy form, which inhibits DNA polymerases and causes chain termination, and, with daunorubicin, is a first-line regimen for AML?

Pharmacogenomic poor metabolizers are at increased risk of mercaptopurine toxicities. What is the involved enzyme?

Summary the mechanisms of the antimetabolites

Folate analogs	Block DHFR, thymidylate synthase and early enzymes in purine synthesis blocks RNA and DNA synthesis; cell cycle arrest
Fluoropyrimidines	Incorporation into RNA blocks RNA synthesis; inhibition of thymidylate synthesis inhibits DNA synthesis; cell cycle arrest
Cytidine analogs	Inhibition of DNA polymerase (by cytarabine): Triphosphorylated deoxycytidine analogs compete with dCTP for incorporation into DNA by DNA polymerase, which blocks elongation of growing DNA strand, leading to DNA breaks with activation of p53 and apoptosis if breaks are not repaired. (Other cytidine analogs have slightly different mechanisms.)
Thiopurines	Enzymatic conversion to thio-purines (toxic nucleotides), which are incorporated in DNA and RNA synthesis, leading to cell cycle arrest in the S phase
Cytidine analogs	Activation to triphosphorylated molecule, which inhibits DNA polymerases and terminates strand elongation and triggers apoptosis by incorporation into the growing DNA strand. Other cytidine analogs have additional actions.

Antitumor Antibiotics

Anthracyclines

*Doxorubicin

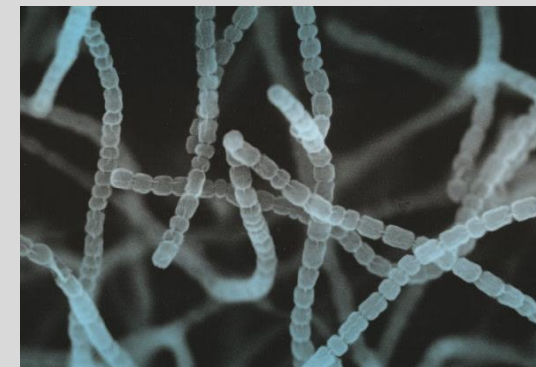
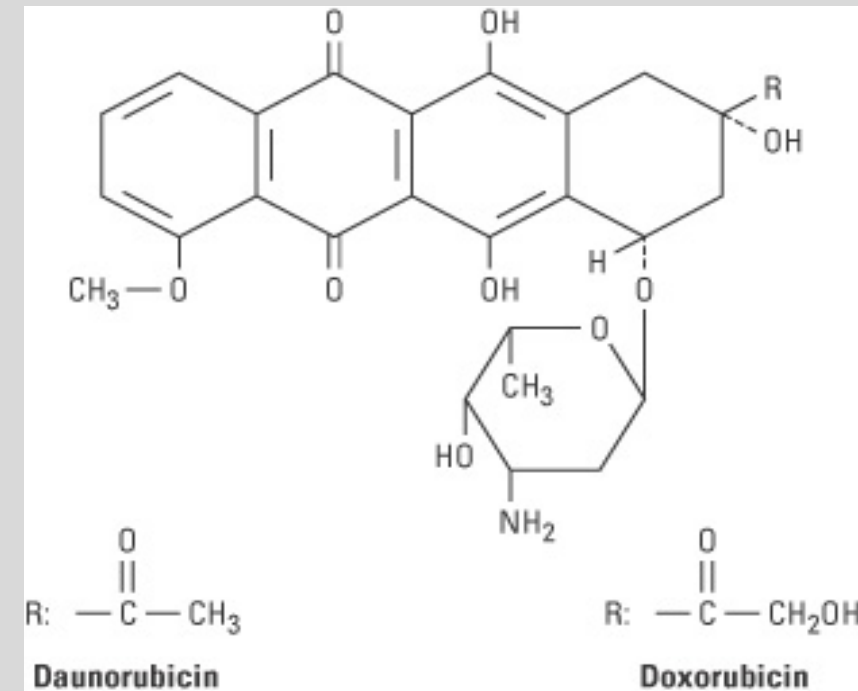
Daunorubicin

Idarubicin

Epirubicin

Valrubicin

Cell cycle: nonspecific



Streptomyces peucetius produces anthracyclines as a secondary metabolite, which is harvested for use as a chemotherapeutic agent.

Other antitumor antibiotics:

Mitoxantrone, Bleomycin, Mitomycin, Dactinomycin

By Docwarhol - Own work, CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=64416332>

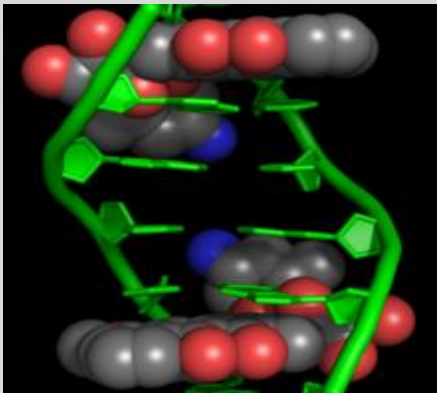
PK Doxorubicin

A	I.V.	
D	Protein binding ~75% Widely distributed Accumulates in tissues	<u>P-glycoprotein substrate</u> Does not cross BBB
M	<u>CYP2D6, 3A4</u>	Inhibits several CYPs including CYP3A4
E	Feces: Biliary excretion, unchanged drug and metabolites	
$t_{1/2}$	Multiphasic; $t_{1/2\text{terminal}}$ range 20 to 48 h (age and gender differences) Pegylated liposomal doxorubicin: ~55 h	
	🔔 Anthracyclines turn urine red	

rubicin → ruby → red



Anthracyclines Mechanisms of Action

Topoisomerase II inhibition	Intercalation in the DNA	Free radical formation
3-way complex: Drug — topo II — DNA complex ↓ Irreparable DNA breaks ↓ apoptosis	Insert between adjacent base pairs ↓ local uncoiling ↓ 1.Blocks DNA and RNA synthesis and repair, and 2.Causes DNA strand scission http://en.wikipedia.org/wiki/Doxorubicin 	Semiquinone and oxygen free radicals (iron-dependent, enzyme-mediated reductive process) H_2O_2 and $-OH$ radicals ↓ attack DNA and oxidize DNA bases ↓ single strand DNA scission + damage to proteins and membranes

Anthracyclines Therapeutic Uses and Resistance mechanisms

Doxorubicin therapeutic uses: Broad spectrum

Wide range of solid tumors, lymphomas, multiple myeloma, sarcomas

Multidrug resistance is common

1. cellular export of drug by MRP transporter family and by the closely related breast cancer resistance protein (BCRP) transporters
2. ↑ glutathione peroxidase activity
3. ↓ activity or mutation of topoisomerase II
4. ↑ repair of DNA strand breaks

MRP: multidrug resistance-associated protein

Anthracyclines Adverse Effects

Cardiotoxicity <i>the most distinctive toxicity</i>	Pericarditis-myocarditis syndrome Chronic: cumulative dose-related, irreversible, progression to cardiomyopathy and acute left ventricular failure <i>Cardiomyopathy is proportional to the cumulative exposure.</i>
Major dose-limiting effects	Myelosuppression (major) Mucositis, stomatitis
Neurotoxicity	Peripheral and CNS
Injection-related	Extravasation / severe local vesicular effects
Radiation recall	Severe local skin reaction in irradiated tissues
Malignancy	Leukemogenic; myelodysplastic syndrome



Source: Wolff K, Johnson RA, Saavedra AP: *Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology*, Seventh Edition: www.accessmedicine.com
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Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Severe tissue necrosis after extravasation of doxorubicin
Fitzpatrick's Dermatology in General Medicine, 8e
Figure 227-9

Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Upper left: Radiation dermatitis: acute, recall phenomenon
This patient had breast cancer. She had a lumpectomy, methotrexate, and x-ray therapy and developed painful erythema and erosions at the irradiated site.

Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, 7e >
Chapter 227. Cytotoxic and Antimetabolic Agents
Figure 10-29

Upper right: Reactivation of solar erythema 2 days after high-dose methotrexate administration to a young woman with choriocarcinoma.

Fitzpatrick's Dermatology in General Medicine, 8e >
Chapter 227. Cytotoxic and Antimetabolic Agents
Figure 227-8

Anthracyclines Drug Interactions

Doxorubicin + Trastuzumab	both cause cardiac toxicity ↑ risk of cardiac toxicity
Paclitaxel + Anthracycline	Paclitaxel competition for CYP3A4 and P-glycoprotein → Paclitaxel increases anthracycline serum concentration ↑ risk of anthracycline toxicity
➤ Practice Point: <i>The anthracycline should be administered prior to the taxane.</i> “A before T”	
Metabolic and transport related	Inducers / inhibitors / substrates of CYP2D6, CYP3A4, P-glycoprotein

Natural Products

Mitotic inhibitors

- Vinca Alkaloids
- Taxanes

Topoisomerase inhibitors

- Camptothecin Analogs
- Epipodophyllotoxins

Vinca Alkaloids:

*Vincristine

Vinblastine

Vinorelbine

(semisynthetic)



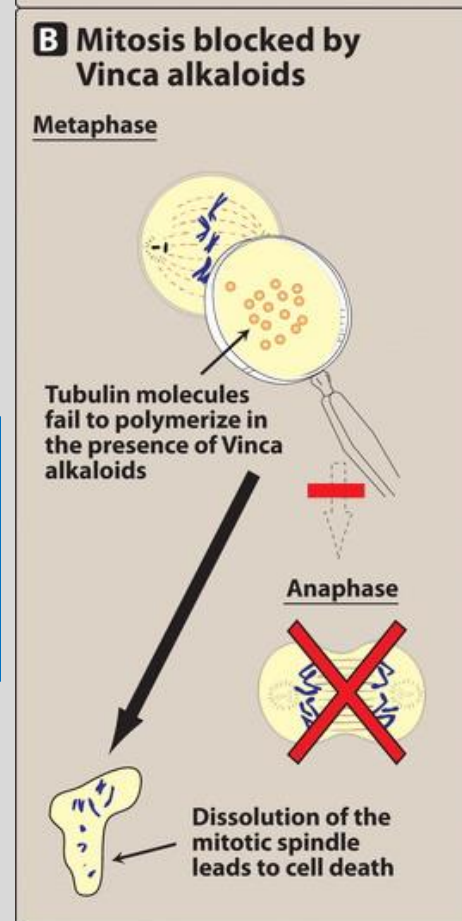
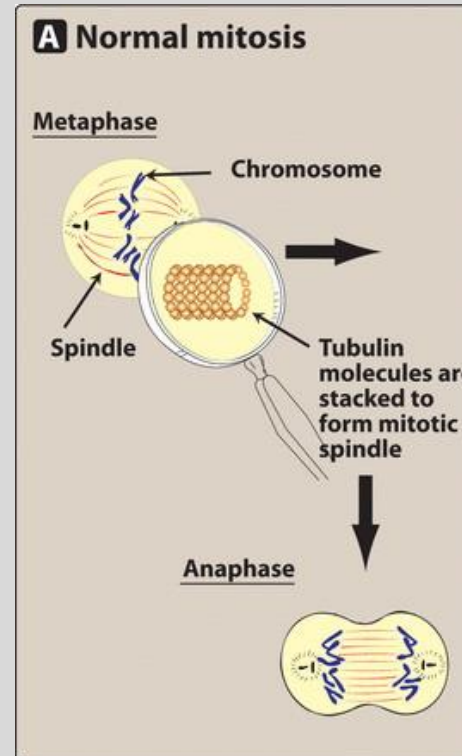
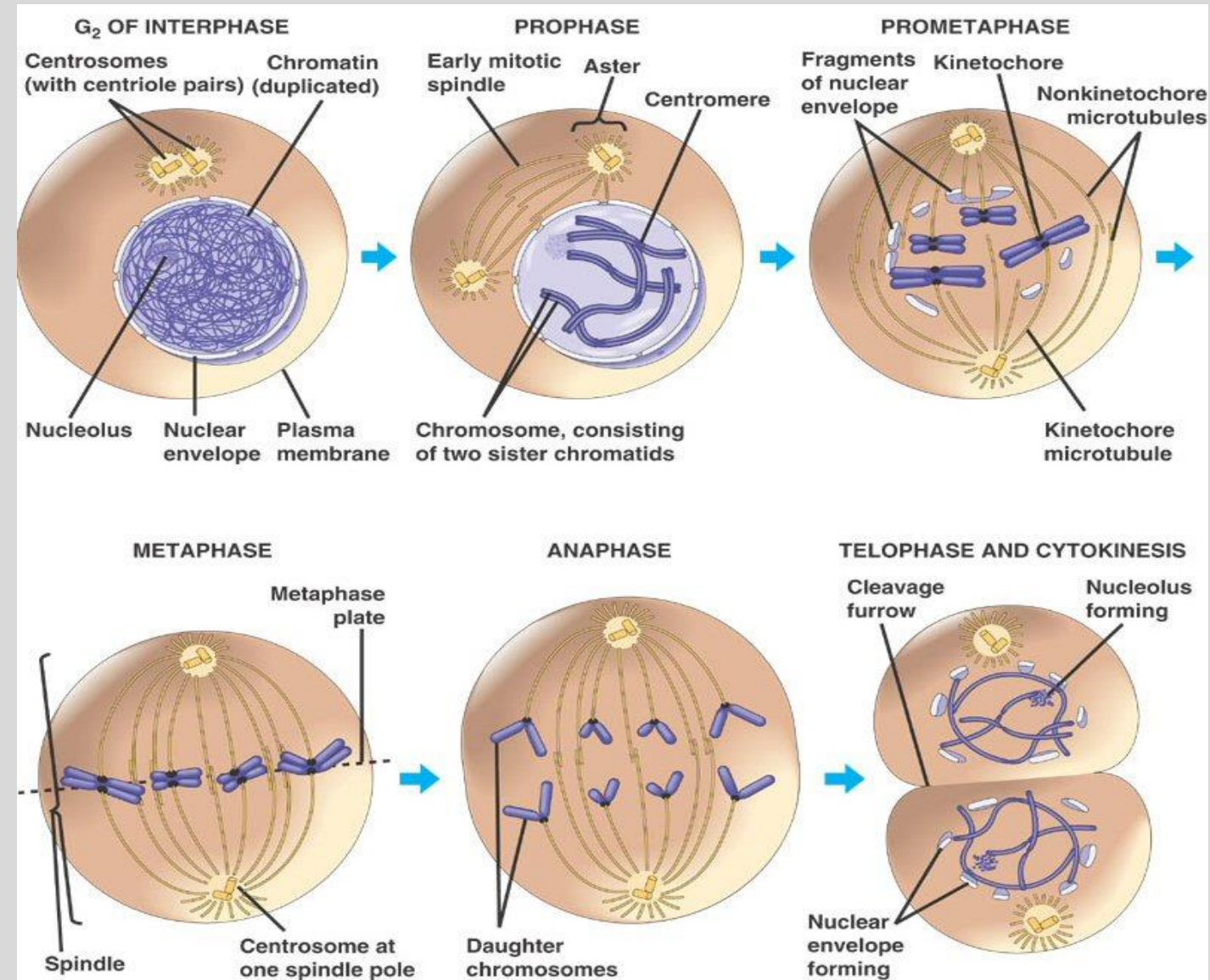
Catharanthus roseus

Madagascar Periwinkle

The leaves are the source of vinca alkaloids,
vinblastine and vincristine

Vincristine pharmacology

PK	IV Pgp substrate, poor penetration of CSF CYP3A4 metab t _{1/2} 20-45 h <i>Intrathecal admin contraindicated</i>
MOA	Inhibition of microtubule polymerization arresting the cell in metaphase (next slide)
Resistance	MDR resistance is common Multidrug resistant tumor cells display cross-resistance to vinca alkaloids, the epipodophyllotoxins, anthracyclines, and taxanes. • Expression of efflux pumps MDR, MRP, BCRP, Pgp • Mutations in β-tubulin or $\downarrow$ expression of the β-tubulin
Uses	Vinca alkaloids have different activities and therapeutic uses Vincristine: Pediatric and adult leukemias, lymphomas, solid tumors ➤ Treatment of choice to induce remission in childhood leukemia
Vincristine toxicities	Neurotoxicity (main dose-limiting toxicity, can be severe) Peripheral neuropathy, autonomic nervous system dysfunction, motor dysfunction Constipation: can be severe with abdominal pain / obstruction Vesicant Extravasation
	CYP3A4 and P-gp related drug interactions



Vinca alkaloids bind to β -tubulin and **block polymerization** with alpha tubulin.

Cells are blocked in mitosis.
(M phase) $\Rightarrow$ Apoptosis

Inhibition of microtubule function results in ***mitotic arrest in metaphase*** and brings cell division to a halt, which then leads to cell death.

Taxanes

*Paclitaxel

Nabpaclitaxel

nanoparticle

albumin-bound paclitaxel

*Docetaxel

Cabazitaxel

Cell division arrest in M phase of cell cycle



Taxus baccata (European yew)

Paclitaxel is isolated from the bark of the Pacific yew.

Docetaxel is derived from needles of the European yew.

Pharmacokinetics: Taxanes

Paclitaxel

- Pgp substrate, extensive protein binding
- Hepatic CYP2C8, CYP3A4 metabolism
- Fecal excretion of metabolites
- $t_{1/2\text{terminal}}$ 13-52 hours

Nonlinear clearance

likely due to ethanol-polyoxyethylated castor oil diluent

Docetaxel

- Linear clearance
- Hepatic CYP3A4 metabolism
- Pgp substrate
- Extensive protein binding (albumin, alpha-1-acid gp, lipoproteins)
- fecal excretion of metabolites
- $t_{1/2\text{terminal}}$ ~11 hours

Nabpaclitaxel:

Nanoparticle albumin bound paclitaxel

Soluble in aqueous solution

Higher serum concentrations / increased clearance

→ similar systemic exposure

Cabazitaxel

Lower affinity for MDR proteins

Hepatic CYP3A4 metabolism

$t_{1/2}$ 95 h

Note: High potential for CYP-mediated drug interactions.

Taxanes bind β -tubulin with high affinity.

High affinity binding to β -tubulin stabilizes microtubules and prevents microtubule disassembly.



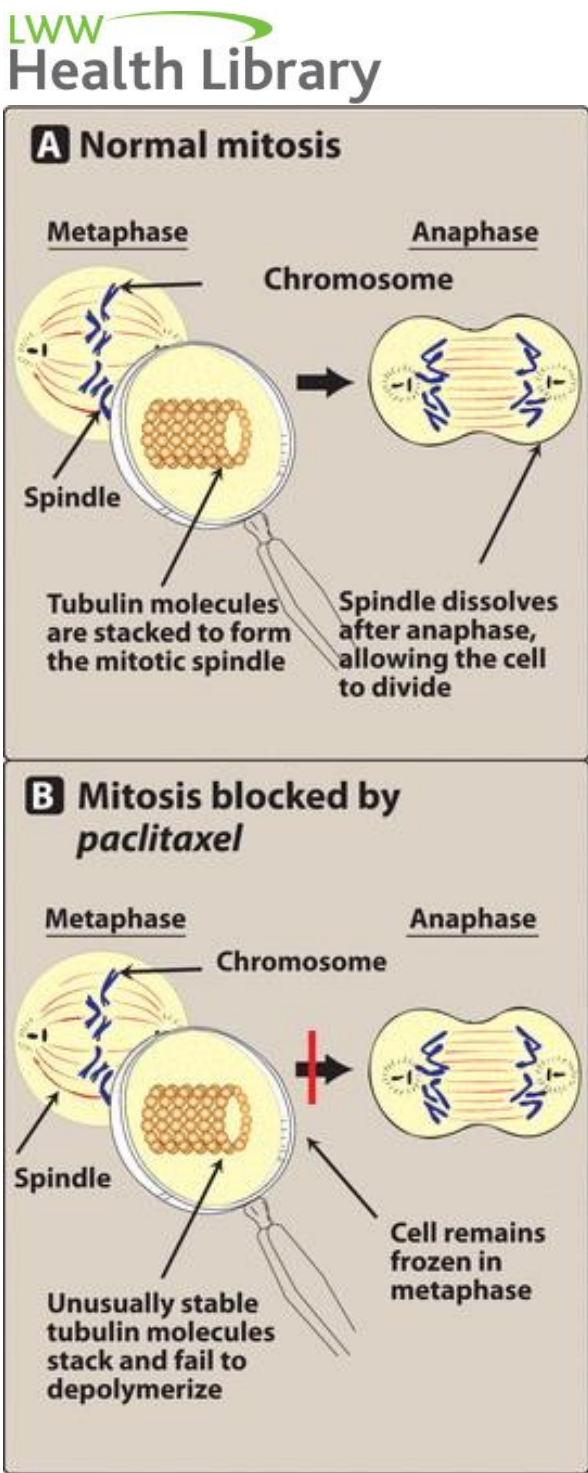
Defects in mitotic spindle assembly, chromosome segregation, and cell division structures derived from microtubules accumulate.



Cell cycle arrest



Apoptosis



Therapeutic Uses

Paclitaxel	Broad spectrum of solid tumors
Nabpaclitaxel	
Docetaxel	
Cabazitaxel	Treatment of castration-resistant metastatic prostate cancer

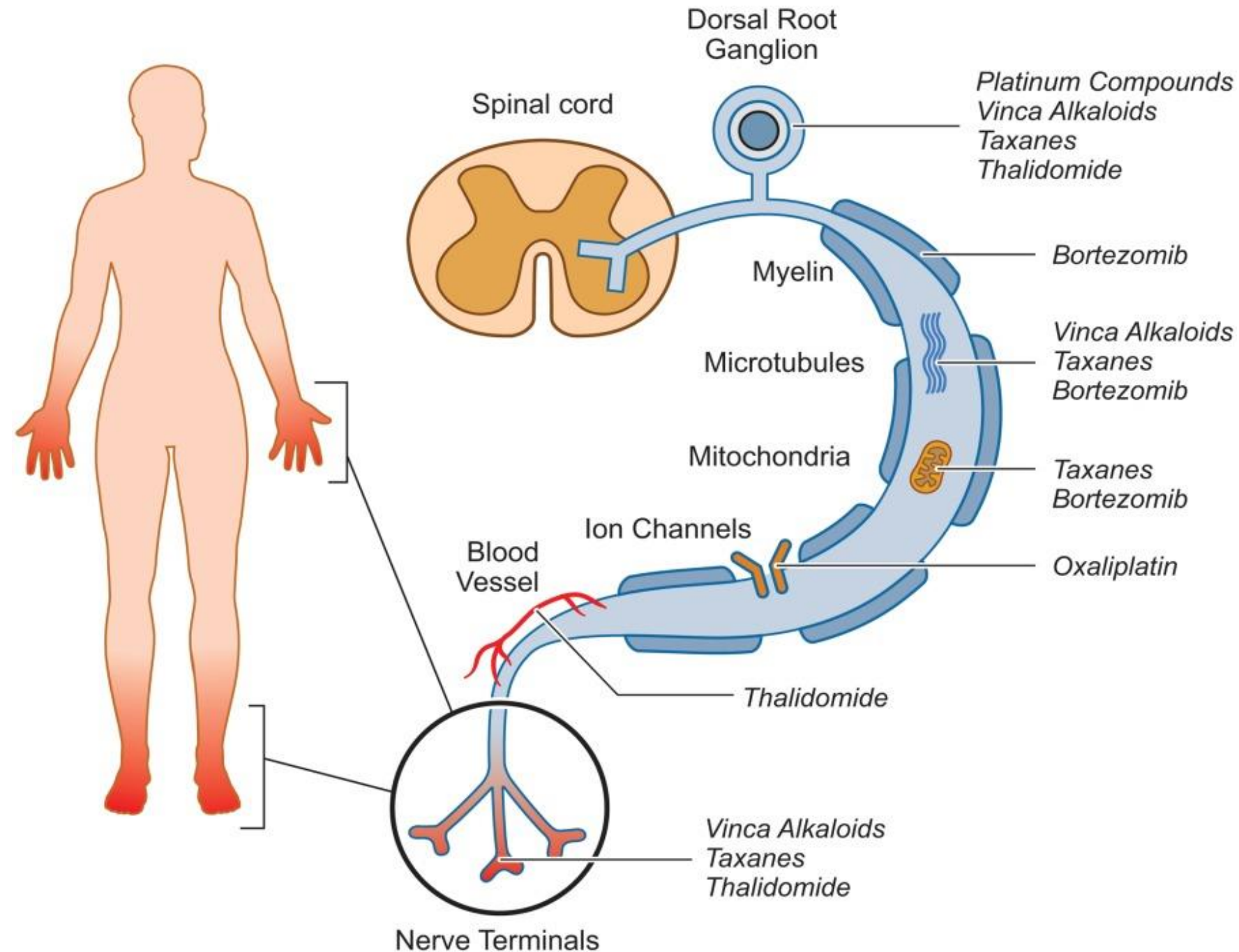
Resistance:

- ↑ Expression of MDR genes
- Mutations in β -tubulin

Toxicities

- Myelosuppression with neutropenia
- Peripheral neuropathy (“glove-stocking”)
- Radiation recall: Severe reaction in irradiated skin
- Fluid retention, ascites, general edema
- Extravasation / severe vesicular effects
- Administer anthracycline before taxane

Chemotherapy-induced peripheral neurotoxicity: A critical analysis



Camptothecin Analogs

*Irinotecan

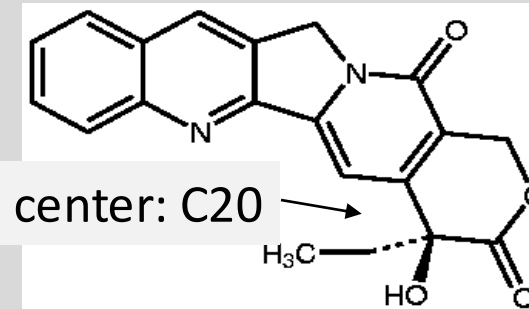
Topotecan

Arrest in S phase of cell cycle

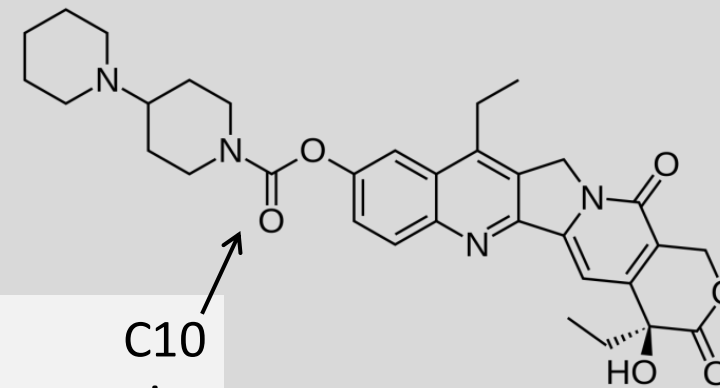
Requirements for biologic activity:

1. S-conformation of the chiral center at C20 in the lactone ring
 2. C10 hydroxyl group
- **Irinotecan** is a prodrug converted to **active SN-38** by carboxylesterase
 - **Topotecan** is not a prodrug

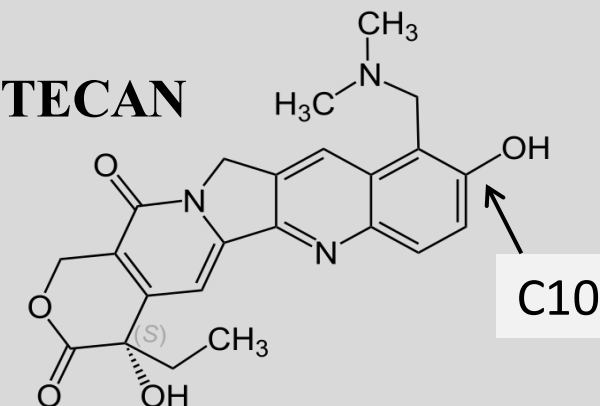
CAMPTOTHECIN



IRINOTECAN



TOPOTECAN



Camptotheca acuminate
Camptothecins are extracted from bark.

Closed-ring form is active. | Open-ring form is inactive.

In biologic fluids, active SN-38 and topotecan lactone are hydrolyzed to **inactive open ring** carboxylate forms.



Active closed-ring lactone ⇌ Inactive open-ring configuration
dynamic equilibrium

Irinotecan	Topotecan
1. Irinotecan ^{carboxyesterase} → <u>active SN-38</u> 2. SN-38 → <u>UGT1A1</u> → glucuronide metabolite (inactive)	No conversion to active lactone ring form • Plasma / hepatic metabolism → inactivate metabolite <i>UGT1A1 is not involved</i>
Pharmacogenomics and Irinotecan: <u>UGT1A1*28</u> → <i>deficient</i> UGT activity → ↑ irinotecan toxicity, especially diarrhea, can be severe. Dose reduction is recommended in patients with the homozygous UGT1A1*28 allele.	

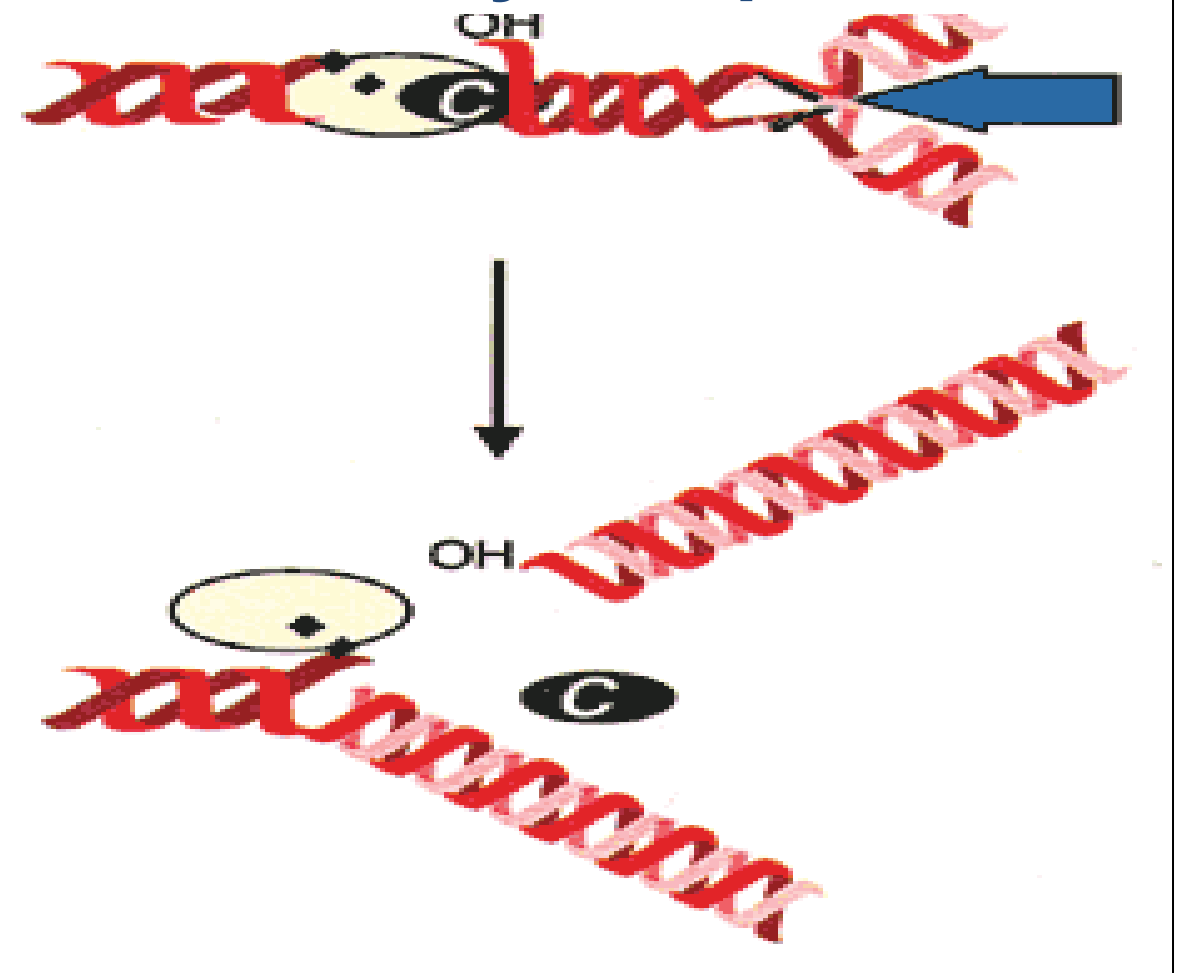
Camptothecins Mechanism of Action

1. Drug binds to topoisomerase I-nicked DNA complex.
2. DNA strand resealing is prevented.
3. Single strand breaks in DNA accumulate.
4. Eventually, the replication fork collides with the complex → irreversible double-strand DNA break → arrest in S phase → cell death.

Mechanisms of Resistance: not well understood

- Drug efflux
- Topoisomerase I alterations
- Other potential mechanisms

Drug-topoisomerase I-DNA ternary complex



<http://www.australianprescriber.com/magazine/25/5/108/10/>

Figure 1 (Black oval with C = Camptothecin)

Camptothecins

Treatment of metastatic and recurrent
and progressive tumors

Irinotecan

Numerous solid tumors
Metastatic colorectal cancer FOLFIRI
Folinic acid – 5-FU – Irinotecan

Topotecan

Recurrent ovarian cancer
Small cell lung cancer
Cervical cancer
Ewing's sarcoma
Rhabdomyosarcoma
CNS malignancy

Adverse effects

Bone marrow suppression – Dose-limiting: Both

Neutropenia can be severe and potentially fatal
especially when used in combination with other
drugs that cause hematological toxicity

Diarrhea – Dose-limiting: Irinotecan

Diarrhea may be severe with
electrolyte disturbances, dehydration, and
complicated by colitis, ulceration, bleeding
Cholinergic sx may occur w. higher doses

Typical adverse effects of chemo drugs

mucositis, nausea, vomiting, pulmonary toxicity,
renal toxicity, etc.

Epipodophyllotoxins

*Etoposide



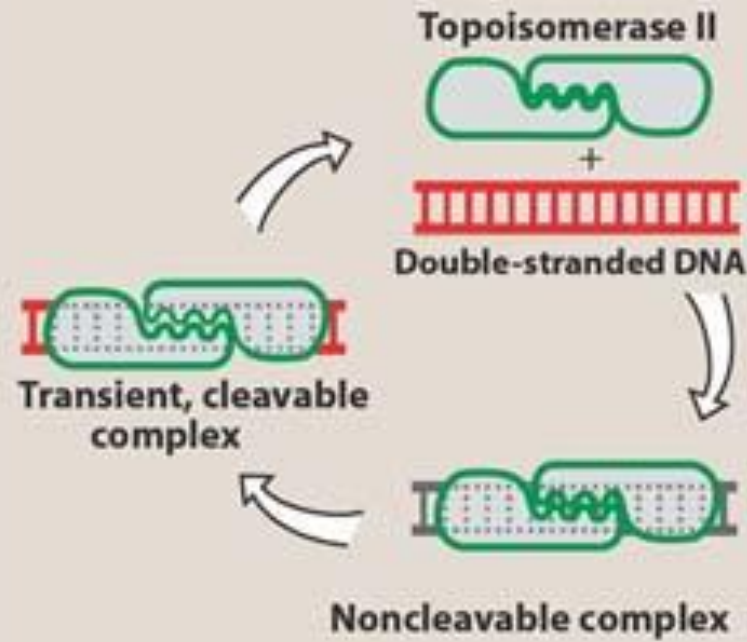
Therapeutic uses	Toxicities
Numerous hematologic cancers and solid tumors, including small cell lung cancer and testicular cancer (with bleomycin and cisplatin)	Myelosuppression Nausea/vomiting Mucositis Alopecia Secondary acute leukemia

Podophyllum peltatum

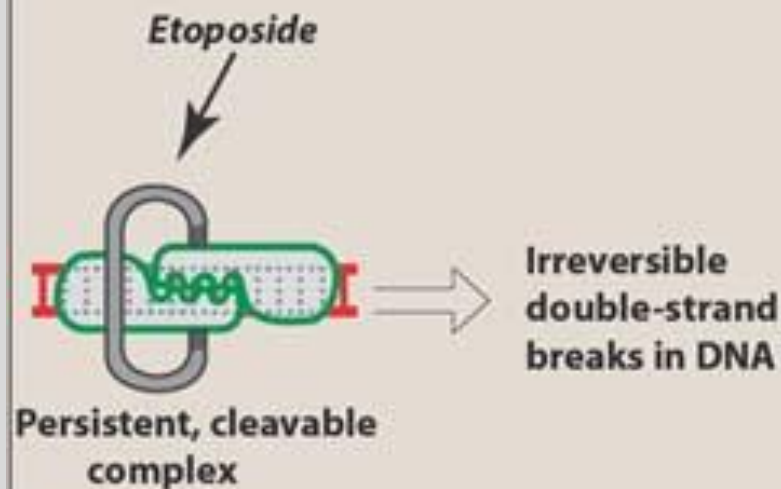
American Mayapple
(Mandrake)

Epipodophyllotoxins are extracted from the roots.

A Normal catalytic cycle
of topoisomerase



B Etoposide leads to double-strand breaks in DNA



Epipodophyllotoxins Mechanisms

1. Formation of a ternary complex
drug-DNA-topoisomerase II complex
2. Inhibition of topoisomerase II → prevents resealing of double-strand breaks
3. The enzyme remains bound to the free end of the broken DNA strand → accumulation of DNA breaks → arrest → apoptosis

Cell cycle arrest in late S and G2 phase

Resistance Mechanisms

1. Efflux:
2. Topoisomerase II alterations
2. Mutations in *p53* tumor suppressor gene
 - P-53 is required for inducing apoptosis

Check your knowledge

A broad spectrum chemotherapy causes irreversible cardiotoxicity that can lead to left ventricular dysfunction, cardiomyopathy, and heart failure, apparently resulting from generation of free radicals and lipid peroxidation. What is the mechanism of action of this drug?

What are the other dose-limiting toxicities of this drug?

A patient with a history of rhabdosarcoma treated with a chemotherapy regimen that includes vincristine continues to suffer with painful pins and needles sensation, motor weakness and muscle cramps in his upper and lower extremities. What is the adverse effect?

Check your knowledge

What is the difference between the mechanisms of action of the vinca alkaloids and the taxanes?

What are the dose-limiting toxicities of each drug in the regimen?

Check your knowledge

A patient with KRAS mutated metastatic colorectal cancer is given the FOLFIRI regimen. What does “IRI” represent and what is its dose-limiting side effect?

What are other dose-limiting toxicities of this drug?

A 28-year-old male diagnosed with testicular cancer is treated with the first-line regimen, BEP regimen. E represents a drug that prevents resealing of the double strand breaks in DNA ultimately leading to cell cycle arrest in S-G2 phase. What is the name of the drug?

Check your knowledge

What are the mechanisms of action of the other two drugs in the BEP regimen?

(I don't expect you to know the answer so I will tell you because it is Boards Relevant.)

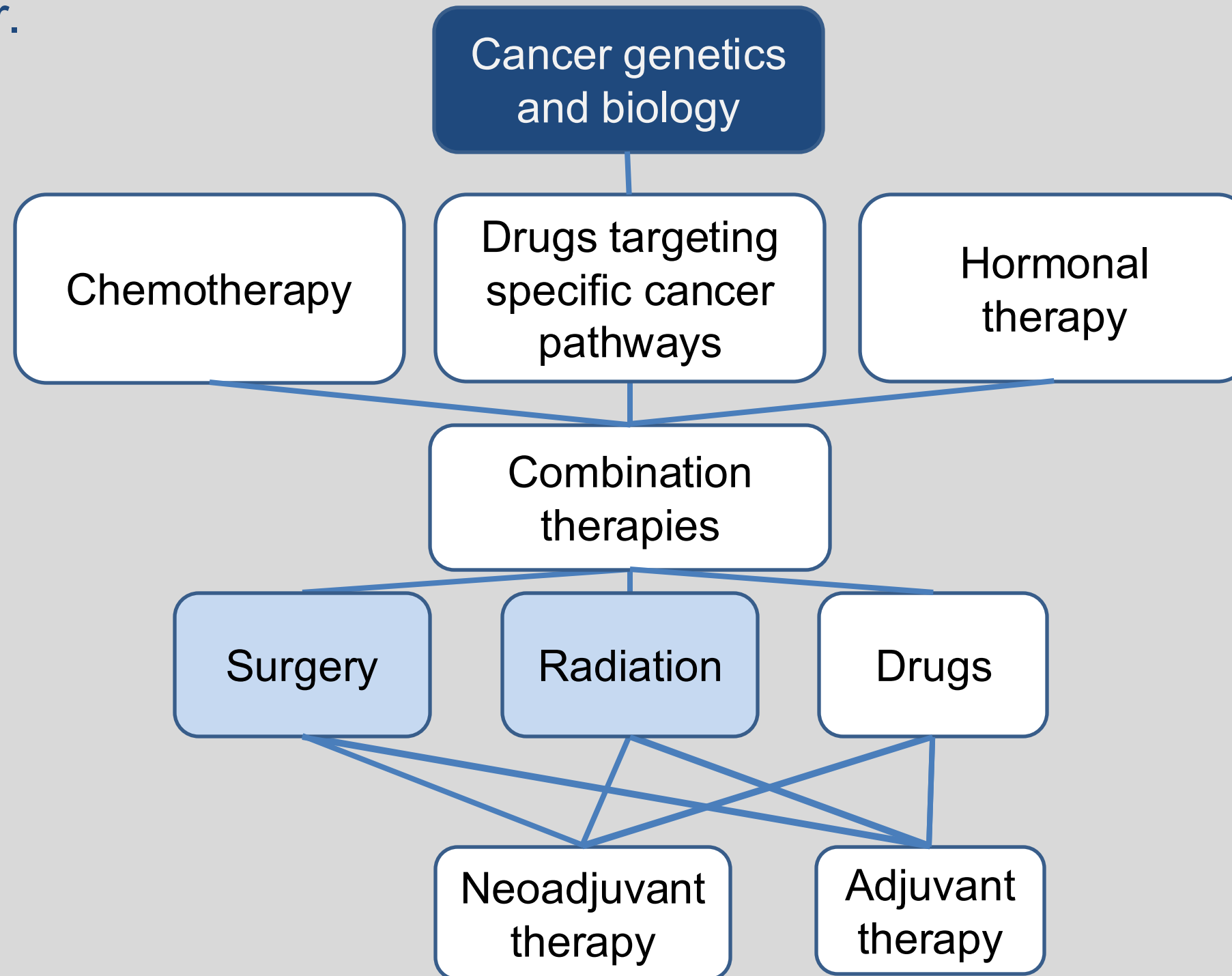
What is an important mechanism of resistance common to the anthracyclines and the antitumor natural products?

Summary:

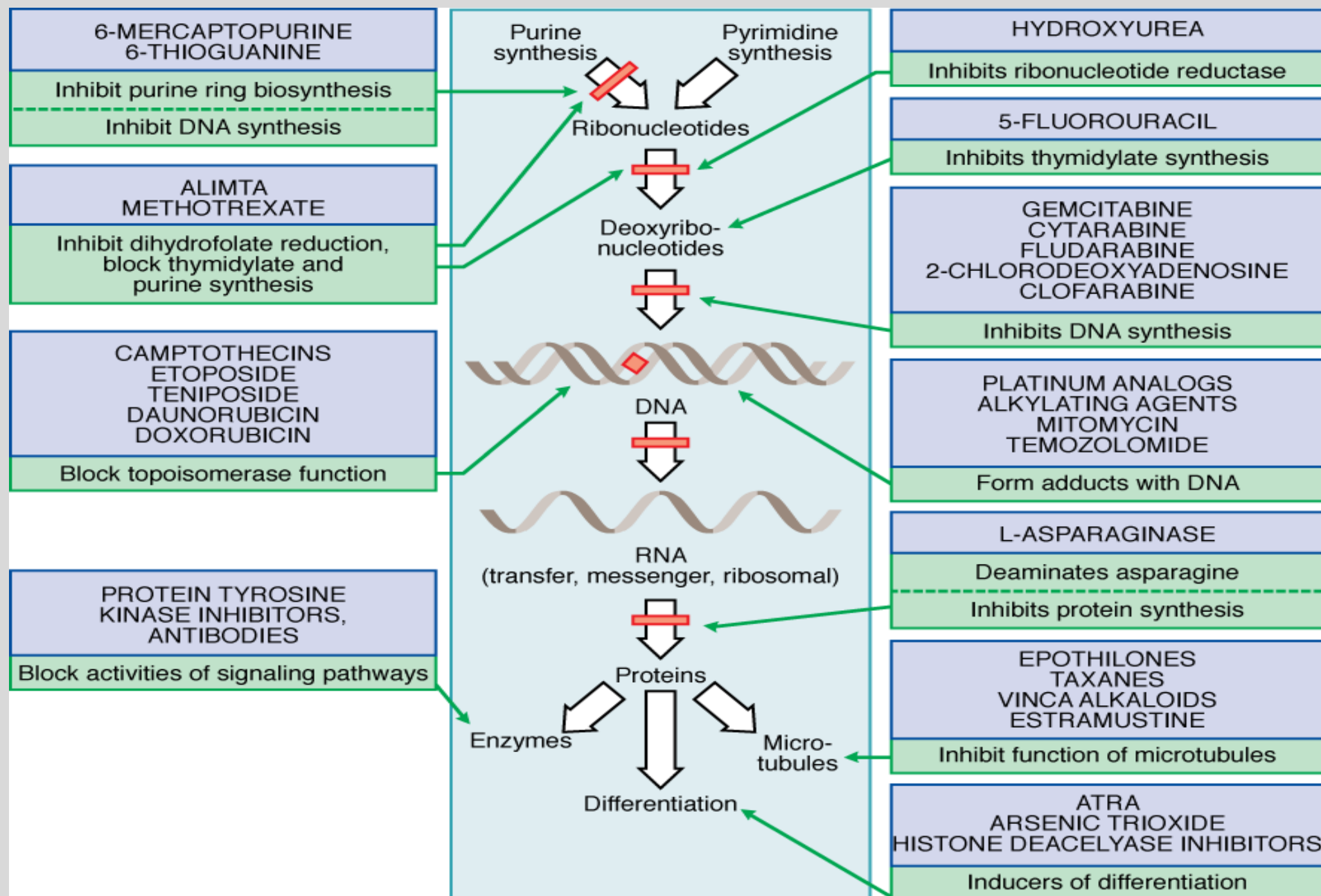
Nonselective Chemotherapy:

Classifications and Mechanisms of Cytotoxicity

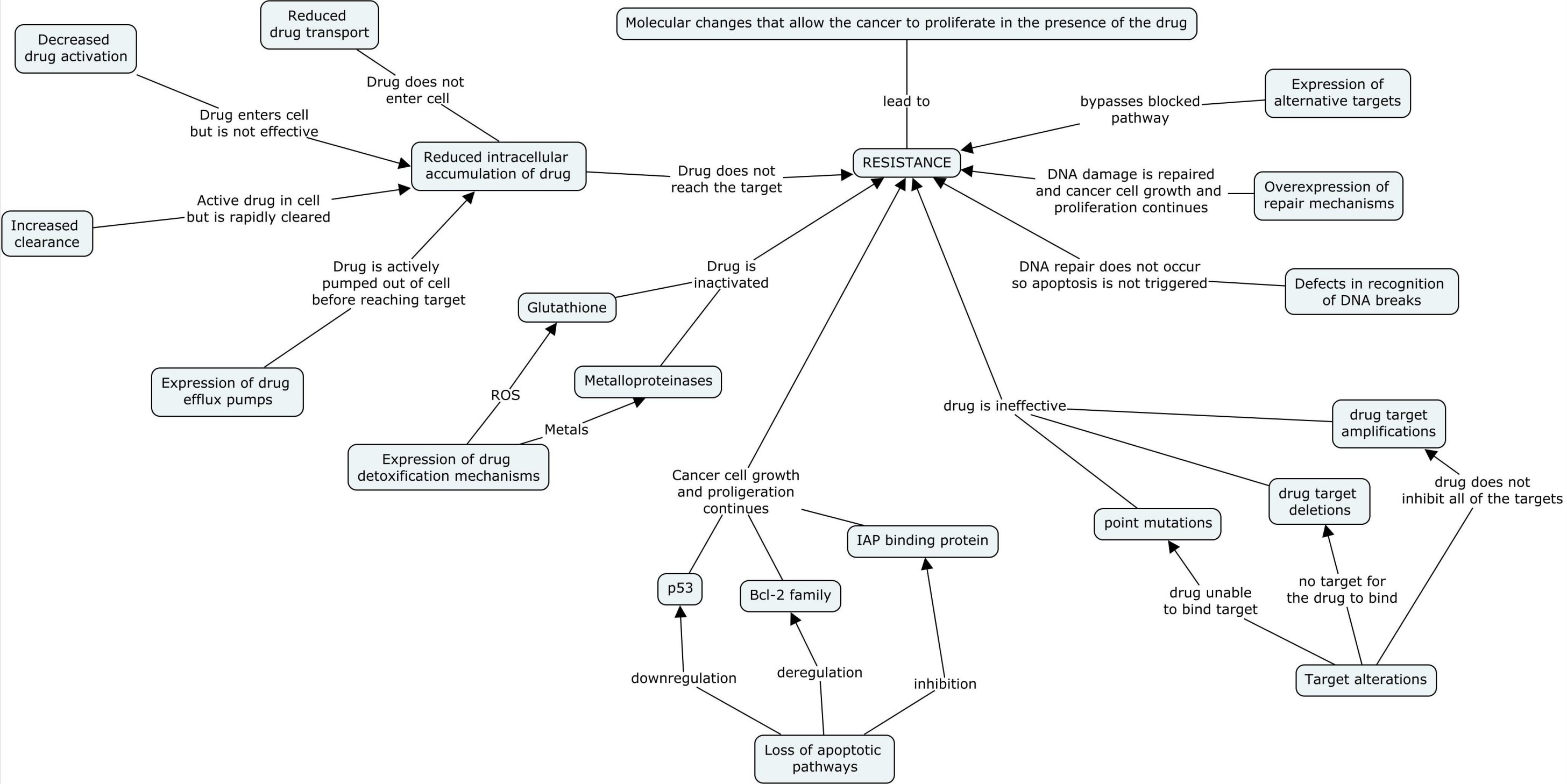
Here is a framework of the main ideas about the pharmacotherapy of cancer.



SITES OF ACTION



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics*, 12th Edition: www.accessmedicine.com
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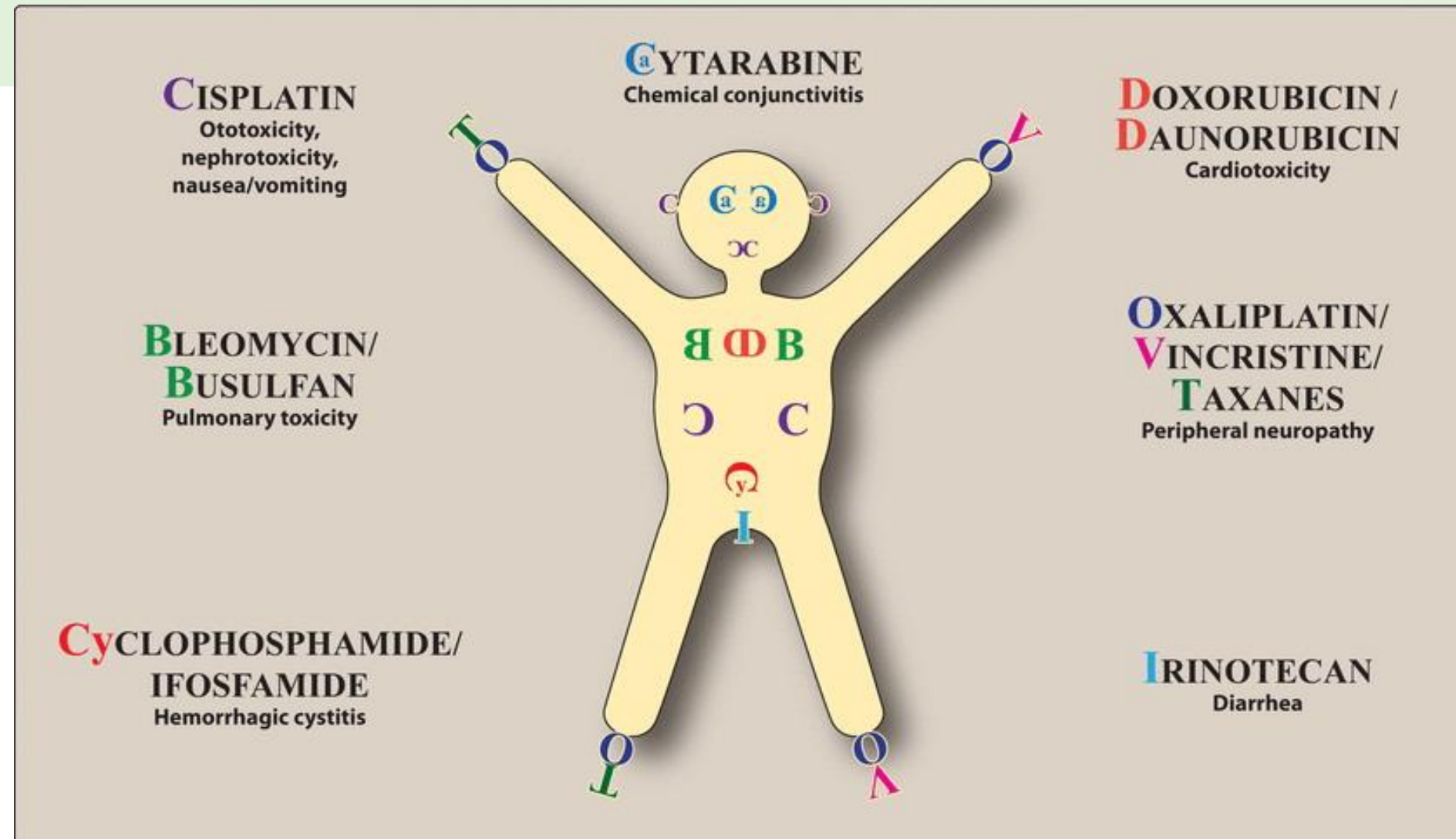
Class / Drugs	Mechanism
Alkylating/ Adducting Agents: Cell cycle nonspecific (occurs at any stage). Cells are most vulnerable in G1 and S phases	
Alkylating agents *Cyclophosphamide Ifosfamide Many others	Transfer alkyl groups N7 position of guanine (mainly) and other points on DNA. Also react with other cellular nucleophiles. Alkylation of DNA results inter- and intrastrand cross-linking, abnormal base pairing (miscoding), scission of DNA backbone, and strand breakage.
Platinum analogs *Cisplatin *Carboplatin *Oxaliplatin	Binds guanine in DNA, forming adducts with inter- and intrastrand cross-linking, which inhibits polymerases, thus inhibiting DNA replication/synthesis and RNA synthesis.

Class / Drugs	Mechanism
Antimetabolites: S phase cell cycle specific activity	
Antifolates: *Methotrexate Two others	Inhibit DHFR and thymidylate synthase, which inhibits de novo purine nucleotide synthesis
Fluoropyrimidines: *5-Fluorouracil *Capecitabine One more	1- 5-FU is converted to 5-FdUMP. Forms stable complex: 5-FdUMP-folate-thymidylate synthase that inhibits thymidylate synthesis 2- Incorporation of FdUTP into DNA leads to strand breaks and faulty DNA replication 3- Incorporation of FUTP in RNA disrupts RNA production and function
Deoxycytidine analogs: *Cytarabine Gemcitabine	Cyt: Converted to active Ara-CTP that inhibits DNA polymerase α , which blocks DNA synthesis, and polymerase β , which blocks DNA repair; AraCTP incorporation into DNA and RNA inhibits their synthesis. Gem: similar, plus inhibition of ribonucleotide reductase
Purine antagonists *6-Mercaptopurine 6-Thioguanine Two others	Converted to toxic purine nucleotides that are incorporated into DNA and RNA inhibiting their synthesis

Antitumor Antibiotics: Cell cycle nonspecific	
Antitumor Antibiotics: *Doxorubicin Four other anthracyclines, Two other antibiotics	1- Topoisomerase II inhibition and irreversible DNA strand scission 2- Intercalation by high-affinity binding to DNA, blocking DNA and RNA synthesis 3- Generation of semiquinone free radicals that cause damage to DNA, proteins, and cell membranes
Microtubule Inhibitors: M phase cell cycle specific in metaphase	
Vinca alkaloids: Vinblastine *Vincristine Vinorelbine	Bind β -tubulin → blocks tubulin polymerization into microtubules → disrupts formation of mitotic spindle → chromosomal segregation and cell proliferation are inhibited
Taxanes: *Paclitaxel *Docetaxel Two others	Bind with high-affinity to microtubules → stabilizes existing microtubules → prevents mitotic spindle function and cell division → cell cycle arrest → apoptosis
Topoisomerase Inhibitors: S phase cell cycle specific	
Camptothecins: *Irinotecan Topotecan	Inhibit topoisomerase I, the enzyme that cuts and religates single DNA strands
Epipodophyllotoxins: *Etoposide	Inhibits topoisomerase II, the enzyme that cuts and re-ligates double stranded DNA

From: **35 Anticancer Drugs**

Lippincott® Illustrated Reviews: Pharmacology, 7e, 2019



Legend:

Chemo Man—a summary of toxicity of chemotherapeutic agents.

Check your knowledge

What class of drugs induces cell death by nucleophilic attack on DNA bases, esp. the N7 guanine, leading to inter- and intrastrand cross-linking, abnormal base pairing, miscoding, and strand breakage.

- The alkylating agents. DNA damage activates of DNA repair systems that create DNA strand breaks by repair enzymes, which activates the apoptosis pathway.

What cellular mechanisms are essential for cell death?

- Recognition of the adduct
- Intact mismatch repair and DNA damage repair systems to induce DNA strand breaks by repair enzymes
- Normal p53 gene and activation of checkpoint proteins
- **Resistance develops rapidly when a drug is used as a single agent**

What is the active cytotoxic metabolite of cyclophosphamide (and ifosfamide)?

- Phosphoramidate mustard: cyclophosphamide → hydroxycyclophosphamide → aldophosphamide → phosphoramidate mustard

Check your knowledge

What is the name of the urotoxic metabolite and how can the toxicity be mitigated?

- Acrolein toxicity to bladder epithelium causes hemorrhagic cystitis.
- Mesna conjugates and inactivates acrolein, decreasing the risk of hemorrhagic cystitis

What is a mechanisms of tumor resistance to cyclophosphamide and ifosphamide?

- Increased expression of aldehyde dehydrogenase increases the rate of metabolism of aldophosphamide to inactive carboxyphosphamide.
- Increased expression glutathione transferase detoxifies alkylating intermediates

Cyclophosphamide is used in the treatment of what categories of cancers?

- Numerous solid and blood tumors and immunosuppression in autoimmune diseases.

List the common and serious toxicities of cyclophosphamide.

- bone marrow suppression, liver tox., cardiotox, lung tox, malignancies and class tox.

Check your knowledge

How does the mechanism of the platinum analogs compare with the mechanism of the alkylating agents?

- Alkylating agents: Nucleophilic attack on DNA → cross-linking → activate DNA repair mechanisms → apoptosis
- Platinum analogs: Diffuse through copper transporter → aquation to positively charged molecules → nucleophilic attack on DNA → cross-linking → activate DNA repair mechanisms → apoptosis. Platinum analogs also bind nuclear and cytoplasmic proteins
- Apoptosis requires that p53 and other checkpoint proteins are intact.

What are the main mechanisms of resistance to platinum analogs?

- Efflux (ATP7A/B copper exporters)
- Inactivation (metalloproteinases, glutathione-S-transferase, sulfhydryls)
- Loss of DNA mismatch repair and decreased apoptosis, and other mechanisms.

Check your knowledge

What is the definition of dose-limiting toxicities?

- Specific adverse effects of a drug that are severe enough to prevent an increase in the dose. They are critical in determining the maximum tolerated dose during clinical trials, especially in oncology.

What are the dose-limiting toxicities of cisplatin, carboplatin, and oxaliplatin?

- Cisplatin: nephrotoxicity; carboplatin: bone marrow suppression; oxaliplatin: peripheral neuropathy, acute and delayed (progressive)

In sequential administration of a platinum analog with paclitaxel or docetaxel, why is the taxane administered first?

- To limit myelosuppression (taxane) and increase efficacy.
- Pharmacokinetic data during clinical trials with paclitaxel administered in sequential infusion with cisplatin demonstrated a decrease in paclitaxel clearance when paclitaxel was administered after cisplatin.
- The mechanism has not been identified.
- Administer paclitaxel before cisplatin, when given as sequential infusions, to limit toxicity. Problems associated with other taxane/platinum combinations are possible, although unsubstantiated. Administering the taxane derivative before the platinum derivative seems prudent.

Check your knowledge

A broad spectrum chemotherapy causes irreversible cardiotoxicity that can lead to left ventricular dysfunction, cardiomyopathy, and heart failure, apparently resulting from generation of free radicals and lipid peroxidation. What is the mechanism of action of this drug?

- 1) DNA intercalation, 2) inhibition of topoisomerase II which inhibits DNA and RNA synthesis, DNA repair, and results in DNA fragmentation, 3) generation of free radicals, which damage DNA, membrane functions.

What are the other dose-limiting toxicities of this drug?

- Bone marrow suppression, mucositis/stomatitis

A patient with a history of rhabdosarcoma treated with a chemotherapy regimen that includes vincristine continues to suffer with painful pins and needles sensation, motor weakness and muscle cramps in his upper and lower extremities. What is the adverse effect?

- Sensorimotor peripheral neuropathy, which is a dose-limiting toxicity of vincristine.

Check your knowledge

What is the difference between the mechanisms of action of the vinca alkaloids and the taxanes?

- Vinca alkaloids inhibit polymerization in microtubules
- Taxanes stabilize microtubules and prevent depolymerization
- Both suppress spindle microtubule function → cell cycle arrest → apoptosis

A patient diagnosed with breast cancer receives paclitaxel intravenous infusion once weekly every three weeks as a component of her adjuvant chemotherapy regimen that includes doxorubicin and cyclophosphamide.

What are the dose-limiting toxicities of each drug in the regimen?

- Paclitaxel: bone marrow suppression and peripheral neuropathy (sensorimotor, glove-stocking)
- Doxorubicin: cardiotoxicity/cardiomyopathy (cumulative dose); bone marrow suppression
- Cyclophosphamide: bone marrow suppression and hemorrhagic cystitis (due to acrolein metabolites), congestive heart failure

Check your knowledge

A patient with KRAS mutated metastatic colorectal cancer is given the FOLFIRI regimen. What does “IRI” represent and what is its dose-limiting side effect?

- IRI = irinotecan, a prodrug converted to toxic SN-38, a topoisomerase I inhibitor prevents resealing of DNA strand, leading to strand breaks and cell cycle arrest in the S phase.
- DNA strand breaks causes cell cycle arrest in S phase.

What are other dose-limiting toxicities of this drug?

- Bone marrow suppression, mucositis/stomatitis

A 28-year-old male diagnosed with testicular cancer is treated with the first-line regimen, BEP regimen. E represents a drug that prevents resealing of the double strand breaks in DNA ultimately leading to cell cycle arrest in S-G2 phase. What is the name of the drug?

- Etoposide inhibits of topo II by forming drug-DNA-topoisomerase II complex, which prevents resealing of double-strand breaks, causing irreversible double-strand breaks.
- Accumulation of DNA double strand breaks causes cell cycle arrest in late S and G2 phases and apoptosis.

Check your knowledge

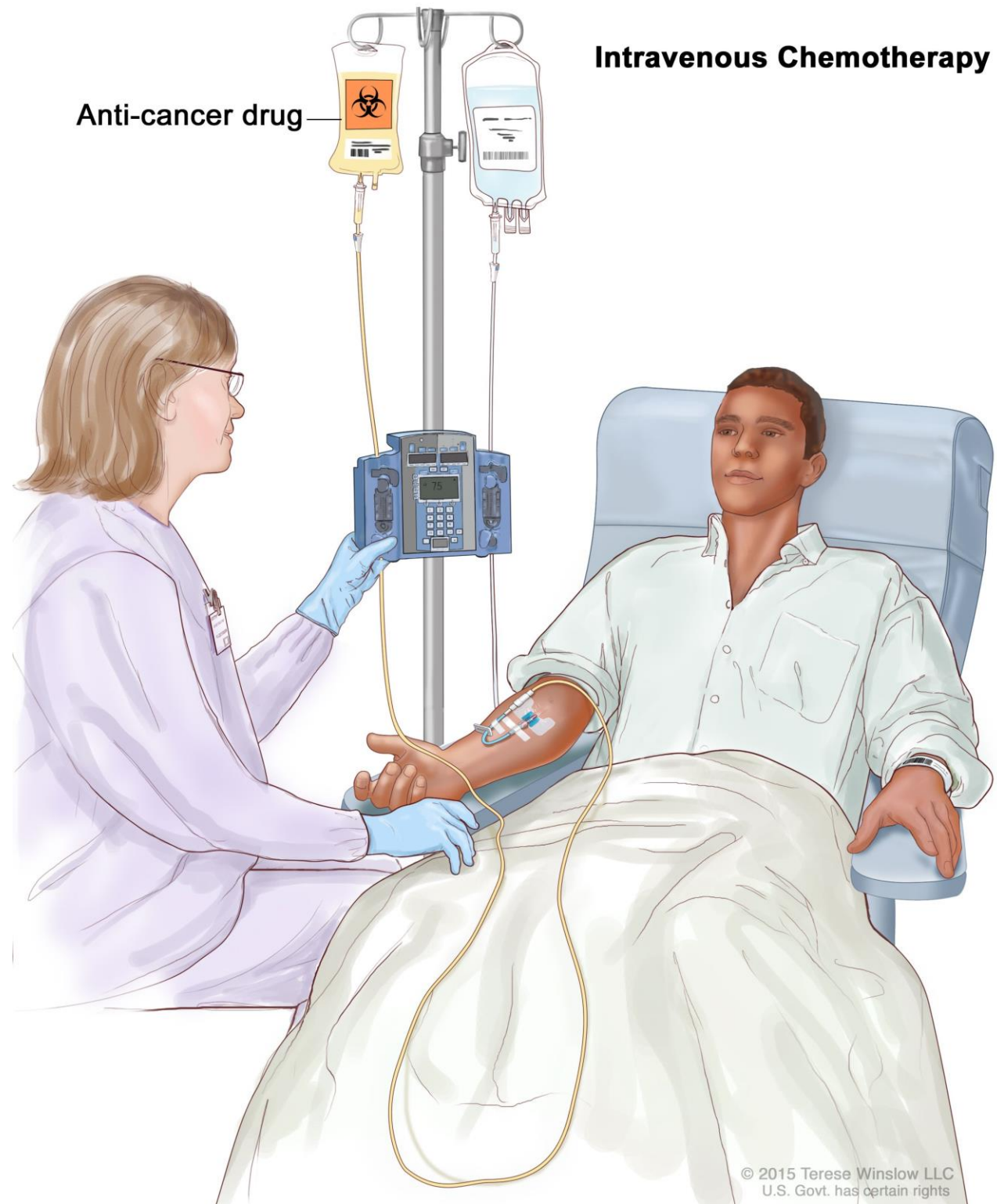
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- B=Bleomycin, an antitumor antibiotic that forms a complex with DNA and iron: DNA-B-Fe²⁺, which is oxidized to DNA-B-Fe³⁺. Superoxide and hydroxyl radicals are formed, which attack phosphodiesterase bonds of DNA, resulting in strand breaks and chromosomal aberrations. Bleomycin causes pulmonary fibrosis (“bleomycin lung”).
- P=platinum=cisplatin. Chlorine molecules on the structure are replaced by water molecules (aquation) in the low-chloride environment inside the tumor cell, which then attacks purine residues on DNA, especially the N⁷ of guanine, leading to inter- and intrastrand breaks and miscoding – apoptosis (if recognized by p53 and other checkpoint proteins). Nephrotoxicity is the dose-limiting toxicity.

What is an important mechanism of resistance common to the anthracyclines and the antitumor natural products?

- Efflux via upregulated expression of efflux pumps: multidrug resistance (MDR) pumps including MDR1= P-glycoprotein, multidrug resistance-associated protein (MRP), breast cancer resistance pump (BCRP).



https://www.physio-pedia.com/Chemotherapy_Side_Effects_and_Syndromes

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